IT5052 – DATA COMMUNICATION & NETWORKING

COMPREHENSIVE MCQ QUESTION BANK & SELF-ASSESSMENT COMPANION

Units IV (Network Layer Protocols), Unit V (Transport Layer Protocols) & Unit VI (Application Layer)

Total Questions: 123 MCQs **Target Exam:** Internal Examination - 2 **Format:** 4 Options + Verified Key

UNIT IV: NETWORK LAYER PROTOCOLS

Q1. Which layer of the OSI model is the Network Layer? [1. Network Layer Introduction]

A) Layer 3 **B)** Layer 2

C) Layer 4 **D)** Layer 7

✓ **Answer: Layer 3**

Q2. What is the primary protocol data unit (PDU) at the Network Layer? [1. Network Layer Introduction]

A) Packet **B)** Frame

C) Segment **D)** Bit

✓ **Answer: Packet**

Q3. Which networking hardware device primarily operates at the Network Layer (Layer 3)? [1. Network Layer Introduction]

A) Router **B)** Repeater

C) Switch (Layer 2) **D)** Hub

✓ **Answer: Router**

Q4. What is the primary responsibility of the Network Layer in computer networking? [1. Network Layer Introduction]

A) Logical addressing, routing, and packet forwarding **B)** Physical synchronization of raw bits

C) Process-to-process communication **D)** Displaying web pages to users

✓ **Answer: Logical addressing, routing, and packet forwarding**

Q5. Which protocol serves as the core and primary protocol of the Network Layer? [1. Network Layer Introduction]

A) IP (Internet Protocol) **B)** TCP

C) HTTP **D)** Ethernet

✓ **Answer: IP (Internet Protocol)**

Q6. Which of the following characteristics best describes the Internet Protocol (IP)? [2. IP - Internet Protocol]

A) Connectionless and best-effort

B) Connection-oriented and reliable

C) Guaranteed ordered delivery

D) Error-free stream delivery

✓ **Answer: Connectionless and best-effort**

Q7. What does 'best-effort delivery' mean in the context of the Internet Protocol (IP)? [2. IP - Internet Protocol]

A) IP does not guarantee delivery, absence of loss, or packet order

B) Packets are delivered only if retransmissions are paid for

C) Packets are encrypted end-to-end automatically

D) Packets always arrive within 1 millisecond

✓ **Answer: IP does not guarantee delivery, absence of loss, or packet order**

Q8. What is the size of an IPv4 address in bits and bytes? [2. IP - Internet Protocol]

A) 32 bits (4 bytes)

B) 64 bits (8 bytes)

C) 128 bits (16 bytes)

D) 16 bits (2 bytes)

✓ **Answer: 32 bits (4 bytes)**

Q9. What is the size of an IPv6 address in bits and bytes? [2. IP - Internet Protocol]

A) 128 bits (16 bytes)

B) 32 bits (4 bytes)

C) 64 bits (8 bytes)

D) 256 bits (32 bytes)

✓ **Answer: 128 bits (16 bytes)**

Q10. Which protocol layer or protocol can be utilized above IP to provide reliability and ordered data delivery? [2. IP - Internet Protocol]

A) TCP at the Transport Layer

B) Ethernet at Data Link Layer

C) UDP at Transport Layer

D) Physical Layer signaling

✓ **Answer: TCP at the Transport Layer**

Q11. What does the acronym ICMP stand for? [3. ICMPv4]

A) Internet Control Message Protocol

B) Internal Computer Monitoring Protocol

C) Inter-network Channel Management Protocol

D) Internet Connection Message Packet

✓ **Answer: Internet Control Message Protocol**

Q12. What is the primary function of ICMPv4 in computer networks? [3. ICMPv4]

A) Sending error reporting messages and diagnostic information

B) Transferring high-resolution multimedia files

C) Assigning permanent MAC addresses to network interfaces

D) Encrypting application payloads

✓ **Answer: Sending error reporting messages and diagnostic information**

Q13. Which popular network diagnostic utility utilizes ICMP Echo Request and Echo Reply messages? [3. ICMPv4]

A) ping
B) telnet
C) ssh
D) netstat

✓ **Answer: ping**

Q14. How are ICMP messages encapsulated and transmitted over networks? [3. ICMPv4]

A) Inside IP datagram packets
B) Directly in physical layer signals
C) Inside TCP segments
D) Inside HTTP responses

✓ **Answer: Inside IP datagram packets**

Q15. Which of the following represents standard error reporting messages in ICMPv4? [3. ICMPv4]

A) Destination Unreachable and Time Exceeded
B) SYN Flood and Reset
C) ACK Timeout and FIN
D) File Not Found and Bad Gateway

✓ **Answer: Destination Unreachable and Time Exceeded**

Q16. What is the fundamental definition of unicast communication? [4. Unicast Routing - Introduction]

A) One sender transmitting data to one specific receiver (One-to-One)
B) One sender transmitting data to all devices on the network
C) One sender transmitting to a selected group
D) Many senders broadcasting simultaneously

✓ **Answer: One sender transmitting data to one specific receiver (One-to-One)**

Q17. What is the primary role of routing in computer networking? [4. Unicast Routing - Introduction]

A) Selecting the optimal path for packets across networks
B) Converting digital signals to analog voltages
C) Encrypting database transactions
D) Allocating memory to operating system processes

✓ **Answer: Selecting the optimal path for packets across networks**

Q18. When a laptop sends an HTTP web request to a specific web server, which communication model is being used? [4. Unicast Routing - Introduction]

A) Unicast
B) Multicast
C) Broadcast
D) Anycast

✓ **Answer: Unicast**

Q19. Which type of routing algorithm shares distance and next-hop vector information exclusively with neighboring routers? [5. Routing Algorithms]

A) Distance Vector routing
B) Link State routing
C) Flooding algorithm
D) Random walk routing

✓ **Answer: Distance Vector routing**

Q20. Which mathematical algorithm is typically used by Distance Vector routing protocols? [5. Routing Algorithms]

A) Bellman-Ford algorithm

B) Dijkstra algorithm

C) Kruskal algorithm

D) Prim algorithm

✓ **Answer: Bellman-Ford algorithm**

Q21. Which type of routing algorithm builds a complete network topology map and calculates shortest paths? [5. Routing Algorithms]

A) Link State routing

B) Distance Vector routing

C) Hop-count vector routing

D) Static default routing

✓ **Answer: Link State routing**

Q22. Which shortest-path algorithm is utilized by Link State routing protocols? [5. Routing Algorithms]

A) Dijkstra algorithm

B) Bellman-Ford algorithm

C) Floyd-Warshall algorithm

D) Binary search algorithm

✓ **Answer: Dijkstra algorithm**

Q23. Which of the following problems is famously associated with Distance Vector routing algorithms? [5. Routing Algorithms]

A) Slow convergence and routing loops (count-to-infinity)

B) Excessive memory consumption due to full topology maps

C) Inability to work with small networks

D) Mandatory requirement of TCP transport

✓ **Answer: Slow convergence and routing loops (count-to-infinity)**

Q24. What does the acronym RIP stand for in computer networking? [6. Unicast Routing Protocols]

A) Routing Information Protocol

B) Remote Internet Protocol

C) Router Interface Program

D) Resource Identification Protocol

✓ **Answer: Routing Information Protocol**

Q25. What routing metric is used by RIP to determine the best path? [6. Unicast Routing Protocols]

A) Hop count

B) Bandwidth and delay

C) Link reliability and MTU

D) Financial link cost

✓ **Answer: Hop count**

Q26. In RIP, what is the maximum number of usable hops, and what does a hop count of 16 indicate? [6. Unicast Routing Protocols]

A) 15 usable hops; 16 indicates unreachable

B) 16 usable hops; 17 indicates unreachable

C) 30 usable hops; 31 indicates unreachable

D) 255 usable hops; 256 indicates unreachable

✓ **Answer: 15 usable hops; 16 indicates unreachable**

Q27. Which transport protocol and port number does RIP utilize for exchanging routing updates? [6. Unicast Routing Protocols]

A) UDP port 520

B) TCP port 179

C) UDP port 69

D) TCP port 80

✓ **Answer: UDP port 520**

Q28. What does the acronym OSPF stand for? [6. Unicast Routing Protocols]

A) Open Shortest Path First

B) Optimal System Packet Forwarding

C) Open Standard Packet Filter

D) Ordered Shortest Path Function

✓ **Answer: Open Shortest Path First**

Q29. Which type of routing algorithm is OSPF based upon? [6. Unicast Routing Protocols]

A) Link State algorithm (Dijkstra)

B) Distance Vector algorithm (Bellman-Ford)

C) Path Vector algorithm

D) Bellman-Moore algorithm

✓ **Answer: Link State algorithm (Dijkstra)**

Q30. How are OSPF packets transported directly over IP? [6. Unicast Routing Protocols]

A) Directly over IP using protocol number 89 (no TCP/UDP header)

B) Over TCP port 89

C) Over UDP port 89

D) Over ICMP messages

✓ **Answer: Directly over IP using protocol number 89 (no TCP/UDP header)**

Q31. What does the acronym BGP stand for, and what is its primary role? [6. Unicast Routing Protocols]

A) Border Gateway Protocol; exchanges routing information between Autonomous Systems

B) Binary Gateway Protocol; routes internal LAN traffic

C) Broadband Group Protocol; distributes broadcast frames

D) Base Grid Protocol; manages private subnets

✓ **Answer: Border Gateway Protocol; exchanges routing information between Autonomous Systems**

Q32. Which transport layer protocol and destination port does BGP use to ensure reliable route exchanges? [6. Unicast Routing Protocols]

A) TCP port 179

B) UDP port 520

C) TCP port 89

D) UDP port 179

✓ **Answer: TCP port 179**

Q33. What is the definition of multicast communication? [7. Multicast Routing & Basics]

A) Transmitting data from one sender to a selected group of receivers (One-to-many selected group)

B) Transmitting data to every device on the physical network

C) Transmitting data from one sender to one receiver

D) Transmitting data from many senders to one receiver

✓ **Answer: Transmitting data from one sender to a selected group of receivers (One-to-many selected group)**

Q34. What is a primary efficiency advantage of multicasting over sending multiple unicast packets? [7. Multicast Routing & Basics]

A) It conserves network bandwidth by duplicating packets only where paths diverge

B) It guarantees zero latency across the Internet

C) It replaces the need for transport layer ports

D) It eliminates router CPU usage entirely

✓ **Answer: It conserves network bandwidth by duplicating packets only where paths diverge**

Q35. Which IPv4 address class is specifically reserved for multicast addressing? [7. Multicast Routing & Basics]

A) Class D

B) Class A

C) Class B

D) Class E

✓ **Answer: Class D**

Q36. What is the defined IPv4 multicast address range? [7. Multicast Routing & Basics]

A) 224.0.0.0 to 239.255.255.255

B) 192.168.0.0 to 192.168.255.255

C) 240.0.0.0 to 255.255.255.254

D) 10.0.0.0 to 10.255.255.255

✓ **Answer: 224.0.0.0 to 239.255.255.255**

Q37. Which protocol is used by IPv4 hosts and local routers to manage multicast group membership? [7. Multicast Routing & Basics]

A) IGMP (Internet Group Management Protocol)

B) ICMP

C) ARP

D) DHCP

✓ **Answer: IGMP (Internet Group Management Protocol)**

Q38. In IPv6, how are all multicast addresses formatted at their start? [7. Multicast Routing & Basics]

A) They begin with FF (e.g. ff00::/8)

B) They begin with FE80

C) They begin with 2001

D) They begin with FC00

✓ **Answer: They begin with FF (e.g. ff00::/8)**

Q39. How is an IPv6 address written and formatted? [9. IPv6 Addressing]

A) 8 groups of hexadecimal digits separated by colons (:)

B) 4 groups of decimal numbers separated by dots (.)

C) 6 groups of octal digits separated by hyphens (-)

D) 16 binary octets separated by commas

✓ **Answer: 8 groups of hexadecimal digits separated by colons (:)**

Q40. Which compression rule is valid when abbreviating an IPv6 address? [9. IPv6 Addressing]

A) Leading zeros in any group can be omitted, and consecutive zero groups can be replaced with :: once

B) Any zero anywhere can be permanently erased without limit

C) Colons can be omitted if letters are present

D) Double colons (::) can be used multiple times in the same address

✓ **Answer: Leading zeros in any group can be omitted, and consecutive zero groups can be replaced with :: once**

Q41. Which transmission mode exists in IPv4 but was completely eliminated in IPv6? [9. IPv6 Addressing]

A) Broadcast

B) Multicast

C) Unicast

D) Anycast

✓ **Answer: Broadcast**

Q42. Which of the following communication types are supported by IPv6? [9. IPv6 Addressing]

A) Unicast, Multicast, and Anycast

B) Unicast, Broadcast, and Simulcast

C) Broadcast and Multicast only

D) Unicast and Broadcast only

✓ **Answer: Unicast, Multicast, and Anycast**

Q43. What is the fixed size of the IPv6 base header? [10. IPv6 Protocol]

A) 40 bytes

B) 20 bytes

C) 32 bytes

D) 64 bytes

✓ **Answer: 40 bytes**

Q44. What is the minimum base header size of an IPv4 packet (without optional fields)? [10. IPv6 Protocol]

A) 20 bytes

B) 40 bytes

C) 8 bytes

D) 16 bytes

✓ **Answer: 20 bytes**

Q45. Why was the header checksum field removed from the IPv6 base header? [10. IPv6 Protocol]

A) To speed up packet processing at intermediate routers since upper/lower layers provide error checks

B) Because IPv6 packets are immune to physical transmission noise

C) Because IPv6 does not allow routers to forward packets

D) To force routers to calculate CRC-32 on payloads

✓ **Answer: To speed up packet processing at intermediate routers since upper/lower layers provide error checks**

Q46. Which protocol does IPv6 utilize instead of the traditional IPv4 ARP (Address Resolution Protocol)? [10. IPv6 Protocol]

A) Neighbor Discovery Protocol (NDP)

B) RARP

C) BOOTP

D) IGMPv3

✓ **Answer: Neighbor Discovery Protocol (NDP)**

Q47. How does packet fragmentation work in IPv6 compared to IPv4? [10. IPv6 Protocol]

A) Routers do not fragment IPv6 packets; fragmentation is performed only by the sending source device

B) Intermediate routers fragment IPv6 packets whenever MTU is exceeded

C) IPv6 packets can never be fragmented under any circumstance

D) The destination host performs fragmentation prior to acknowledgment

✓ **Answer: Routers do not fragment IPv6 packets; fragmentation is performed only by the sending source device**

Q48. Which control and diagnostics protocol accompanies IPv6? [10. IPv6 Protocol]

A) ICMPv6

B) ICMPv4

C) IGMP

D) SNMPv1

✓ **Answer: ICMPv6**

UNIT V: TRANSPORT LAYER PROTOCOLS

Q49. Which layer of the OSI reference model is the Transport Layer? [1. Transport Layer Introduction]

A) Layer 4

B) Layer 3

C) Layer 5

D) Layer 7

✓ **Answer: Layer 4**

Q50. What fundamental type of communication is provided by the Transport Layer? [1. Transport Layer Introduction]

A) Process-to-process communication between applications on different devices

B) Node-to-node hop transmission between adjacent interfaces

C) Bit synchronization across physical cables

D) Host-to-host network packet routing

✓ **Answer: Process-to-process communication between applications on different devices**

Q51. What is the general name of the Protocol Data Unit (PDU) at the Transport Layer? [1. Transport Layer Introduction]

A) Segment (for TCP) or User Datagram (for UDP)

B) Frame

C) Packet

D) Bit stream

✓ **Answer: Segment (for TCP) or User Datagram (for UDP)**

Q52. What identifier is used at the Transport Layer to distinguish between multiple applications running on the same host? [2. Transport Layer Services]

A) Port number

B) IP address

C) MAC address

D) URL path

✓ **Answer: Port number**

Q53. What is the difference between an IP address and a Port number? [2. Transport Layer Services]

A) IP address identifies a host/interface on a network; Port number identifies a specific application/process on that host

B) IP address identifies an application; Port number identifies the router

C) Both are interchangeable 32-bit identifiers

D) Port numbers are assigned by ISPs, whereas IP addresses are embedded in hardware

✓ **Answer: IP address identifies a host/interface on a network; Port number identifies a specific application/process on that host**

Q54. Which of the following is a standard well-known port example? [2. Transport Layer Services]

A) Port 443 for HTTPS web traffic

B) Port 9999 for IPv6 routing

C) Port 25 for DNS queries

D) Port 80 for SSH remote login

✓ **Answer: Port 443 for HTTPS web traffic**

Q55. What does the combination of an IP address and a Port number form? [2. Transport Layer Services]

A) A Socket address

B) A MAC frame

C) A Subnet mask

D) A DNS record

✓ **Answer: A Socket address**

Q56. What are the three primary transport layer protocols in the modern Internet protocol suite? [3. Transport Layer Protocols Overview]

A) UDP, TCP, and SCTP

B) IP, ICMP, and IGMP

C) HTTP, FTP, and SMTP

D) Ethernet, Wi-Fi, and Token Ring

✓ **Answer: UDP, TCP, and SCTP**

Q57. Which protocol is connectionless and lightweight with minimal overhead? [3. Transport Layer Protocols Overview]

A) UDP (User Datagram Protocol)

B) TCP (Transmission Control Protocol)

C) SCTP

D) BGP

✓ **Answer: UDP (User Datagram Protocol)**

Q58. Which protocol provides reliable, connection-oriented byte stream transmission? [3. Transport Layer Protocols Overview]

A) TCP (Transmission Control Protocol)

B) UDP

C) IP

D) ICMP

✓ **Answer: TCP (Transmission Control Protocol)**

Q59. Which protocol is reliable and message-oriented, supporting multi-homing and multi-streaming? [3. Transport Layer Protocols Overview]

A) SCTP (Stream Control Transmission Protocol)

B) UDP

C) TCP

D) OSPF

✓ **Answer: SCTP (Stream Control Transmission Protocol)**

Q60. What does the acronym UDP stand for? [4. UDP]

A) User Datagram Protocol

B) Unified Data Packet

C) Universal Delivery Protocol

D) Unchecked Datagram Program

✓ **Answer: User Datagram Protocol**

Q61. What is the size of the standard UDP header? [4. UDP]

A) 8 bytes

B) 20 bytes

C) 40 bytes

D) 16 bytes

✓ **Answer: 8 bytes**

Q62. Which fields constitute the 8-byte UDP header? [4. UDP]

A) Source Port, Destination Port, Length, and Checksum

B) Sequence Number, Acknowledgment Number, Window Size, and Flags

C) Version, TTL, Protocol, and Checksum

D) IP Source, IP Destination, MAC Source, and MAC Destination

✓ **Answer: Source Port, Destination Port, Length, and Checksum**

Q63. Why is UDP chosen for applications like online gaming, video conferencing, and DNS queries? [4. UDP]

A) It has very low overhead and introduces no connection setup or retransmission delays

B) It encrypts packets automatically with zero CPU overhead

C) It guarantees 100% error-free delivery

D) It reserves dedicated optical bandwidth on routers

✓ **Answer: It has very low overhead and introduces no connection setup or retransmission delays**

Q64. Does UDP provide guaranteed packet order or retransmission of lost datagrams? [4. UDP]

A) No, UDP itself provides neither delivery guarantee nor retransmission

B) Yes, UDP automatically retransmits missing datagrams three times

C) Yes, UDP reorders packets using sequence numbers

D) Only when running on Gigabit Ethernet

✓ **Answer: No, UDP itself provides neither delivery guarantee nor retransmission**

Q65. What does the acronym TCP stand for? [5. TCP]

A) Transmission Control Protocol

B) Transfer Communication Packet

C) Terminal Connection Protocol

D) Transport Control Packet

✓ **Answer: Transmission Control Protocol**

Q66. What is the minimum header size of a TCP segment (without options)? [5. TCP]

A) 20 bytes

B) 8 bytes

C) 40 bytes

D) 16 bytes

✓ **Answer: 20 bytes**

Q67. How does TCP achieve reliable and ordered delivery of application data? [5. TCP]

A) By using sequence numbers, positive acknowledgments, and retransmission timers

B) By broadcasting all segments to every connected device

C) By preventing packet loss at the physical layer

D) By relying on the underlying IP layer for guaranteed delivery

✓ **Answer: By using sequence numbers, positive acknowledgments, and retransmission timers**

Q68. What is the sequence of steps in the standard TCP Three-Way Handshake for connection establishment? [5. TCP]

A) SYN -> SYN-ACK -> ACK

B) ACK -> SYN -> SYN-ACK

C) SYN -> ACK -> DATA

D) HELLO -> ACK -> CONNECT

✓ **Answer: SYN -> SYN-ACK -> ACK**

Q69. What occurs in step 1 of the TCP three-way handshake? [5. TCP]

A) The client sends a SYN packet to request connection synchronization

B) The server responds with an ACK

C) The client sends payload data immediately

D) The connection is abruptly terminated

✓ **Answer: The client sends a SYN packet to request connection synchronization**

Q70. What occurs in step 2 of the TCP three-way handshake? [5. TCP]

A) The server accepts the connection request and responds with a SYN-ACK packet

B) The client sends an ACK acknowledgment

C) The server closes the socket immediately

D) The client and server exchange symmetric encryption keys

✓ **Answer: The server accepts the connection request and responds with a SYN-ACK packet**

Q71. What occurs in step 3 of the TCP three-way handshake? [5. TCP]

A) The client confirms receipt with an ACK packet, completing connection setup

B) The server resets the connection with RST

C) The client sends a FIN request

D) The server retransmits the SYN packet

✓ **Answer: The client confirms receipt with an ACK packet, completing connection setup**

Q72. Which control mechanisms are natively built into TCP to manage network transmission speeds? [5. TCP]

A) Flow control (Sliding Window) and Congestion Control

B) Frequency division multiplexing

C) Spread spectrum modulation

D) Hardware interrupt arbitration

✓ **Answer: Flow control (Sliding Window) and Congestion Control**

Q73. Which flag in the TCP header is used to gracefully close/terminate an established connection? [5. TCP]

A) FIN

B) SYN

C) URG

D) PSH

✓ **Answer: FIN**

Q74. What does the acronym SCTP stand for? [6. SCTP]

A) Stream Control Transmission Protocol

B) Synchronous Communication Transfer Protocol

C) Socket Connection Transport Protocol

D) Simple Control Transmission Packet

✓ **Answer: Stream Control Transmission Protocol**

Q75. Which feature allows SCTP to handle multiple independent message streams within a single connection? [6. SCTP]

A) Multi-streaming
B) Multi-homing
C) Subnetting
D) Frequency hopping

✓ **Answer: Multi-streaming**

Q76. Which feature in SCTP allows an endpoint to bind to multiple IP addresses for link redundancy and failover? [6. SCTP]

A) Multi-homing
B) Multi-streaming
C) Broadcasting
D) Sliding window

✓ **Answer: Multi-homing**

Q77. What term does SCTP use to refer to an established connection between two endpoints? [6. SCTP]

A) Association
B) Socket pair
C) Virtual circuit
D) Pipe

✓ **Answer: Association**

Q78. How does SCTP transmit data in terms of framing? [6. SCTP]

A) Message-oriented (preserves message boundaries)
B) Continuous unformatted byte stream
C) Raw bit frames only
D) Encrypted blocks only

✓ **Answer: Message-oriented (preserves message boundaries)**

UNIT VI: APPLICATION LAYER

Q79. Which layer of the OSI model is the Application Layer? [1. Application Layer Introduction]

A) Layer 7 (highest layer)
B) Layer 1
C) Layer 4
D) Layer 5

✓ **Answer: Layer 7 (highest layer)**

Q80. What is the main role of the Application Layer in computer networks? [1. Application Layer Introduction]

A) To provide network services and interfaces directly to user applications
B) To route packets through intermediate transit routers
C) To manage physical voltage levels on copper wires
D) To regulate node-to-node MAC addresses

✓ **Answer: To provide network services and interfaces directly to user applications**

Q81. Which of the following is a collection of common Application Layer protocols? [1. Application Layer Introduction]

A) HTTP, FTP, SMTP, DNS, and SSH

B) IP, ICMP, ARP, and RARP

C) TCP, UDP, and SCTP

D) Ethernet, PPP, and HDLC

✓ **Answer: HTTP, FTP, SMTP, DNS, and SSH**

Q82. In the Client-Server networking architecture, which role does the client play? [2. Client-Server Programming]

A) The client initiates communication by sending a request for a service

B) The client waits passively and responds to server requests

C) The client controls the physical network routing tables

D) The client assigns MAC addresses to all connecting machines

✓ **Answer: The client initiates communication by sending a request for a service**

Q83. In the Client-Server model, which role does the server play? [2. Client-Server Programming]

A) The server continuously listens for requests, processes them, and returns responses

B) The server initiates connections to random internet clients

C) The server fragments physical layer frames

D) The server terminates local computer power

✓ **Answer: The server continuously listens for requests, processes them, and returns responses**

Q84. Can a single server simultaneously communicate with and serve multiple clients? [2. Client-Server Programming]

A) Yes, a server can handle multiple concurrent client connections

B) No, a server can only communicate with exactly one client at a time

C) Only if all clients share the same IP address

D) Only if client and server run the identical operating system version

✓ **Answer: Yes, a server can handle multiple concurrent client connections**

Q85. Are the Internet and the World Wide Web (WWW) the same thing? [3. WWW and HTTP]

A) No: the Internet is the underlying network infrastructure; the WWW is a service/collection of resources accessed over the Internet

B) Yes: both terms are 100% identical and interchangeable technical terms

C) No: the WWW is physical cabling, while the Internet is software only

D) Yes: the Internet was renamed to WWW in 1999

✓ **Answer: No: the Internet is the underlying network infrastructure; the WWW is a service/collection of resources accessed over the Internet**

Q86. What does the acronym HTTP stand for? [3. WWW and HTTP]

A) Hypertext Transfer Protocol

B) High-speed Text Transmission Protocol

C) Host Terminal Transfer Packet

D) Hyperlink Text Translation Program

✓ **Answer: Hypertext Transfer Protocol**

Q87. Why is HTTP classified as a 'stateless' protocol? [3. WWW and HTTP]

A) Because the server retains no memory of previous client requests across requests

B) Because it cannot run on computers that have hard drives

C) Because it transmits state governments data only

D) Because connection states are stored on DNS root servers

✓ **Answer: Because the server retains no memory of previous client requests across requests**

Q88. What is the standard default port number for unencrypted HTTP traffic? [3. WWW and HTTP]

A) Port 80

B) Port 443

C) Port 21

D) Port 25

✓ **Answer: Port 80**

Q89. What is the standard default port number for secure, encrypted HTTPS traffic? [3. WWW and HTTP]

A) Port 443

B) Port 80

C) Port 22

D) Port 53

✓ **Answer: Port 443**

Q90. What security protocol does HTTPS use to provide encrypted communication between web browser and server? [3. WWW and HTTP]

A) TLS (Transport Layer Security) / SSL

B) IPsec only

C) WPA3

D) WEP

✓ **Answer: TLS (Transport Layer Security) / SSL**

Q91. What does the acronym FTP stand for? [4. FTP]

A) File Transfer Protocol

B) Fast Transmission Packet

C) Folder Tracking Program

D) Format Translation Protocol

✓ **Answer: File Transfer Protocol**

Q92. Which underlying transport protocol does FTP use to guarantee reliable file transfers? [4. FTP]

A) TCP

B) UDP

C) ICMP

D) IPsec

✓ **Answer: TCP**

Q93. FTP famously uses two separate connections. Which port is commonly used for the control connection? [4. FTP]

A) Port 21

B) Port 20

C) Port 80

D) Port 25

✓ **Answer: Port 21**

Q94. In traditional active-mode FTP, which port number is commonly used for the data transfer connection? [4. FTP]

A) Port 20

B) Port 21

C) Port 22

D) Port 110

✓ **Answer: Port 20**

Q95. What is the difference between FTP's control connection and data connection? [4. FTP]

A) Control connection (port 21) sends commands and replies; data connection (port 20) transfers actual file contents

B) Control transfers files; data transfers user passwords

C) Control connection uses UDP; data connection uses TCP

D) There is no difference; both transfer random binary packets

✓ **Answer: Control connection (port 21) sends commands and replies; data connection (port 20) transfers actual file contents**

Q96. What is the primary role of the Simple Mail Transfer Protocol (SMTP)? [5. Electronic Mail]

A) Sending and relaying email messages from client to server or between mail servers

B) Downloading email messages to offline clients

C) Browsing web mail pages

D) Translating domain names into IP addresses

✓ **Answer: Sending and relaying email messages from client to server or between mail servers**

Q97. Which protocol is traditionally used by email clients to retrieve and download emails from a mail server to a local device? [5. Electronic Mail]

A) POP3 (Post Office Protocol version 3)

B) SMTP

C) SNMP

D) SSH

✓ **Answer: POP3 (Post Office Protocol version 3)**

Q98. Which email retrieval protocol allows users to access, organize into folders, and synchronize emails directly on the mail server? [5. Electronic Mail]

A) IMAP (Internet Message Access Protocol)

B) POP3

C) SMTP

D) FTP

✓ **Answer: IMAP (Internet Message Access Protocol)**

Q99. What are the common standard port numbers associated with SMTP? [5. Electronic Mail]

A) Port 25 (traditional), 587 (submission), and 465 (secure)

B) Port 80 and 443

C) Port 20 and 21

D) Port 110 and 995

✓ **Answer: Port 25 (traditional), 587 (submission), and 465 (secure)**

Q100. What are the standard default port numbers for POP3 (unencrypted and secure)? [5. Electronic Mail]

A) Port 110 (standard) and Port 995 (secure SSL/TLS)

B) Port 143 and Port 993

C) Port 25 and Port 465

D) Port 21 and Port 990

✓ **Answer: Port 110 (standard) and Port 995 (secure SSL/TLS)**

Q101. What are the standard default port numbers for IMAP (unencrypted and secure)? [5. Electronic Mail]

A) Port 143 (standard) and Port 993 (secure SSL/TLS)

B) Port 110 and Port 995

C) Port 25 and Port 587

D) Port 80 and Port 443

✓ **Answer: Port 143 (standard) and Port 993 (secure SSL/TLS)**

Q102. What does the acronym SSH stand for? [6. SSH]

A) Secure Shell

B) System Security Host

C) Socket Secure Handler

D) Server Sharing Header

✓ **Answer: Secure Shell**

Q103. What is the primary function of SSH in network administration? [6. SSH]

A) Providing encrypted remote login and command-line execution over insecure networks

B) Sending audio-video broadcasts to multiple computers

C) Assigning DHCP dynamic IP addresses

D) Filtering spam emails on mail servers

✓ **Answer: Providing encrypted remote login and command-line execution over insecure networks**

Q104. What is the default TCP port number used by SSH servers? [6. SSH]

A) Port 22

B) Port 23

C) Port 21

D) Port 25

✓ **Answer: Port 22**

Q105. Which legacy, insecure protocol does SSH commonly replace for remote command-line access? [6. SSH]

A) Telnet (which transmits passwords in plaintext)

B) FTP

C) DNS

D) HTTPS

✓ **Answer: Telnet (which transmits passwords in plaintext)**

Q106. What does the acronym DNS stand for, and what is its main purpose? [7. DNS]

A) Domain Name System; translates human-friendly domain names into IP addresses

B) Data Network Service; compresses video packets

C) Dynamic Node Synchronization; assigns MAC addresses

D) Digital Name Server; encrypts email text

✓ **Answer: Domain Name System; translates human-friendly domain names into IP addresses**

Q107. What is the standard port number used by DNS servers? [7. DNS]

A) Port 53

B) Port 25

C) Port 80

D) Port 110

✓ **Answer: Port 53**

Q108. Which transport layer protocols can DNS utilize? [7. DNS]

A) Both UDP and TCP (UDP for standard queries; TCP for zone transfers and large responses)

B) UDP only

C) TCP only

D) ICMP only

✓ **Answer: Both UDP and TCP (UDP for standard queries; TCP for zone transfers and large responses)**

Q109. Which DNS record type maps a domain name to a 32-bit IPv4 address? [7. DNS]

A) A record

B) AAAA record

C) MX record

D) CNAME record

✓ **Answer: A record**

Q110. Which DNS record type maps a domain name to a 128-bit IPv6 address? [7. DNS]

A) AAAA record

B) A record

C) NS record

D) TXT record

✓ **Answer: AAAA record**

Q111. Which DNS record type specifies the mail server responsible for receiving emails on behalf of a domain? [7. DNS]

A) MX record

B) CNAME record

C) PTR record

D) A record

✓ **Answer: MX record**

Q112. Which DNS record type creates an alias pointing one domain name to another canonical domain name? [7. DNS]

A) CNAME record

B) A record

C) NS record

D) SOA record

✓ **Answer: CNAME record**

Q113. Which DNS record type delegates a DNS zone to use specific authoritative name servers? [7. DNS]

A) NS record

B) A record

C) MX record

D) AAAA record

✓ **Answer: NS record**

Q114. What is the primary objective of data compression in computer networks? [8. Data Compression]

A) To reduce data size to save storage space and transmission time

B) To encrypt plain text to prevent eavesdropping

C) To convert binary data into human-readable characters

D) To assign unique port numbers to applications

✓ **Answer: To reduce data size to save storage space and transmission time**

Q115. What is the defining characteristic of Lossless compression? [8. Data Compression]

A) No information is lost; the original data can be reconstructed perfectly

B) Some information is permanently discarded to maximize size reduction

C) It only compresses audio files

D) Data cannot be decompressed once compressed

✓ **Answer: No information is lost; the original data can be reconstructed perfectly**

Q116. Which of the following are prime examples of Lossless compression formats? [8. Data Compression]

A) ZIP and PNG

B) JPEG and MP3

C) MPEG video and MP3 audio

D) Lossy WebP audio

✓ **Answer: ZIP and PNG**

Q117. What is the defining characteristic of Lossy compression? [8. Data Compression]

A) Some information is permanently removed, achieving higher compression ratios where exact precision is not critical

B) Original data can always be reconstructed bit-for-bit

C) It is used exclusively for text source code and database backups

D) It never reduces file size

✓ **Answer: Some information is permanently removed, achieving higher compression ratios where exact precision is not critical**

Q118. Which of the following are prime examples of Lossy compression formats? [8. Data Compression]

A) JPEG and MP3

B) ZIP and 7z

C) PNG and FLAC

D) TAR and GZ

✓ **Answer: JPEG and MP3**

UNIT IV: NETWORK LAYER PROTOCOLS

Q119. Which comparison between IPv4 and IPv6 is correct regarding address format? [Comparison: IPv4 vs IPv6]

A) IPv4 uses decimal dotted notation; IPv6 uses hexadecimal colon notation

B) IPv4 uses colons; IPv6 uses decimal dots

C) Both use dotted hexadecimal notation

D) IPv4 is 128-bit; IPv6 is 32-bit

✓ **Answer: IPv4 uses decimal dotted notation; IPv6 uses hexadecimal colon notation**

Q120. Which feature is present in the IPv4 base header but omitted in the IPv6 base header to increase router forwarding performance? [Comparison: IPv4 vs IPv6]

A) Header Checksum

B) Hop Limit

C) Source Address

D) Payload Length

✓ **Answer: Header Checksum**

Q121. Which pair correctly matches the routing protocol with its underlying algorithm? [Comparison: Routing Protocols]

A) RIP -> Bellman-Ford; OSPF -> Dijkstra

B) RIP -> Dijkstra; OSPF -> Bellman-Ford

C) BGP -> Dijkstra; RIP -> Prim

D) OSPF -> Bellman-Ford; BGP -> Floyd

✓ **Answer: RIP -> Bellman-Ford; OSPF -> Dijkstra**

UNIT V: TRANSPORT LAYER PROTOCOLS

Q122. Which comparison correctly pairs transport protocol header sizes? [Comparison: TCP vs UDP]

A) TCP minimum header is 20 bytes; UDP header is 8 bytes

B) TCP header is 8 bytes; UDP header is 20 bytes

C) Both headers are exactly 40 bytes

D) TCP header is 4 bytes; UDP header is 16 bytes

✓ **Answer: TCP minimum header is 20 bytes; UDP header is 8 bytes**

UNIT VI: APPLICATION LAYER

Q123. Which statement accurately distinguishes SMTP from POP3 and IMAP? [Comparison: Email Protocols]

A) SMTP is used to push/send email; POP3 and IMAP are used to retrieve/access email

B) SMTP retrieves emails; POP3 sends emails

C) SMTP operates on port 110; POP3 operates on port 25

D) IMAP is connectionless; SMTP is connection-oriented

✓ **Answer: SMTP is used to push/send email; POP3 and IMAP are used to retrieve/access email**

UNIT IV: NETWORK LAYER PROTOCOLS

Q124. What is the full form of IP? [Abbreviations & Full Forms]

A) Internet Protocol

B) Internal Protocol

C) Interface Program

D) Inter-network Packet

✓ **Answer: Internet Protocol**

Q125. What is the full form of ICMP? [Abbreviations & Full Forms]

A) Internet Control Message Protocol

B) Inter-network Connection Management Protocol

C) Internal Communication Message Packet

D) Internet Channel Monitoring Protocol

✓ **Answer: Internet Control Message Protocol**

Q126. What is the full form of RIP? [Abbreviations & Full Forms]

A) Routing Information Protocol

B) Router Interface Program

C) Remote Internet Protocol

D) Resource Identification Packet

✓ **Answer: Routing Information Protocol**

Q127. What is the full form of OSPF? [Abbreviations & Full Forms]

A) Open Shortest Path First

B) Optimal System Packet Forwarding

C) Open Standard Packet Filter

D) Ordered Shortest Path Function

✓ **Answer: Open Shortest Path First**

Q128. What is the full form of BGP? [Abbreviations & Full Forms]

A) Border Gateway Protocol

B) Binary Gateway Protocol

C) Broadband Group Protocol

D) Base Grid Protocol

✓ **Answer: Border Gateway Protocol**

Q129. What is the full form of IGMP? [Abbreviations & Full Forms]

A) Internet Group Management Protocol

B) Internal Gateway Monitoring Protocol

C) Inter-network Group Messaging Protocol

D) Interface Group Multicast Packet

✓ **Answer: Internet Group Management Protocol**

Q130. What is the full form of ARP? [Abbreviations & Full Forms]

A) Address Resolution Protocol

B) Application Routing Protocol

C) Asynchronous Relay Packet

D) Automated Resource Program

✓ **Answer: Address Resolution Protocol**

Q131. What is the full form of NDP in IPv6? [Abbreviations & Full Forms]

A) Neighbor Discovery Protocol

B) Network Datagram Protocol

C) Node Distribution Protocol

D) Next-hop Delivery Packet

✓ **Answer: Neighbor Discovery Protocol**

Q132. What is the full form of MTU? [Abbreviations & Full Forms]

A) Maximum Transmission Unit

B) Minimum Transport Unit

C) Modular Translation Utility

D) Multi-packet Transmission Unit

✓ **Answer: Maximum Transmission Unit**

Q133. What is the full form of AS in routing? [Abbreviations & Full Forms]

A) Autonomous System

B) Automated Switch

C) Application Server

D) Address Space

✓ **Answer: Autonomous System**

UNIT V: TRANSPORT LAYER PROTOCOLS

Q134. What is the full form of UDP? [Abbreviations & Full Forms]

A) User Datagram Protocol
B) Unified Data Packet
C) Universal Delivery Protocol
D) Unchecked Datagram Program

✓ **Answer: User Datagram Protocol**

Q135. What is the full form of TCP? [Abbreviations & Full Forms]

A) Transmission Control Protocol
B) Transfer Communication Packet
C) Terminal Connection Protocol
D) Transport Control Packet

✓ **Answer: Transmission Control Protocol**

Q136. What is the full form of SCTP? [Abbreviations & Full Forms]

A) Stream Control Transmission Protocol
B) Synchronous Communication Transfer Protocol
C) Socket Connection Transport Protocol
D) Simple Control Transmission Packet

✓ **Answer: Stream Control Transmission Protocol**

Q137. What does the SYN flag in TCP stand for? [Abbreviations & Full Forms]

A) Synchronize (Synchronize Sequence Numbers)
B) Syntax
C) System Node
D) Synchronous Network

✓ **Answer: Synchronize (Synchronize Sequence Numbers)**

Q138. What does the ACK flag in TCP stand for? [Abbreviations & Full Forms]

A) Acknowledgment
B) Access Key
C) Active Connection
D) Address Check

✓ **Answer: Acknowledgment**

Q139. What does the FIN flag in TCP stand for? [Abbreviations & Full Forms]

A) Finish (terminate connection)
B) Finalized packet
C) Filter index
D) Flow indication

✓ **Answer: Finish (terminate connection)**

Q140. What does the RST flag in TCP stand for? [Abbreviations & Full Forms]

A) Reset (reset the connection)
B) Restart session
C) Routing status
D) Receiver state

✓ **Answer: Reset (reset the connection)**

UNIT VI: APPLICATION LAYER

Q141. What is the full form of WWW? [Abbreviations & Full Forms]

A) World Wide Web
B) Wide World Web
C) World Wide Windows
D) Web World Wide

✓ **Answer: World Wide Web**

Q142. What is the full form of HTTP? [Abbreviations & Full Forms]

A) Hypertext Transfer Protocol
B) High-speed Text Transmission Protocol
C) Host Terminal Transfer Packet
D) Hyperlink Text Translation Program

✓ **Answer: Hypertext Transfer Protocol**

Q143. What is the full form of HTTPS? [Abbreviations & Full Forms]

A) Hypertext Transfer Protocol Secure
B) High-level Transfer Protocol Standard
C) Hypertext Translation Program Socket
D) Host Terminal Transport Security

✓ **Answer: Hypertext Transfer Protocol Secure**

Q144. What is the full form of TLS? [Abbreviations & Full Forms]

A) Transport Layer Security
B) Terminal Link System
C) Transaction Level Security
D) Transmission Layer Socket

✓ **Answer: Transport Layer Security**

Q145. What is the full form of SSL? [Abbreviations & Full Forms]

A) Secure Sockets Layer
B) Standard Socket Link
C) System Security Level
D) Secure Synchronous Line

✓ **Answer: Secure Sockets Layer**

Q146. What is the full form of FTP? [Abbreviations & Full Forms]

A) File Transfer Protocol
B) Fast Transmission Packet
C) Folder Tracking Program
D) Format Translation Protocol

✓ **Answer: File Transfer Protocol**

Q147. What is the full form of SMTP? [Abbreviations & Full Forms]

A) Simple Mail Transfer Protocol
B) Standard Message Transmission Packet
C) System Mail Transport Program
D) Secure Mail Transition Protocol

✓ **Answer: Simple Mail Transfer Protocol**

Q148. What is the full form of POP3? [Abbreviations & Full Forms]

A) Post Office Protocol version 3
B) Packet Order Protocol 3
C) Public Office Program 3
D) Private Online Protocol 3

✓ **Answer: Post Office Protocol version 3**

Q149. What is the full form of IMAP? [Abbreviations & Full Forms]

A) Internet Message Access Protocol
B) Internal Mail Application Program
C) Interface Mail Access Packet
D) Instant Message Action Protocol

✓ **Answer: Internet Message Access Protocol**

Q150. What is the full form of SSH? [Abbreviations & Full Forms]

A) Secure Shell
B) System Security Host
C) Socket Secure Handler
D) Server Sharing Header

✓ **Answer: Secure Shell**

Q151. What is the full form of DNS? [Abbreviations & Full Forms]

A) Domain Name System
B) Dynamic Network Service
C) Data Node Synchronization
D) Digital Name Server

✓ **Answer: Domain Name System**

Q152. What is the full form of URL? [Abbreviations & Full Forms]

A) Uniform Resource Locator
B) Universal Resource Link
C) Unified Reference Location
D) Universal Routing Line

✓ **Answer: Uniform Resource Locator**

Q153. What is the standard port number for DNS queries? [Well-Known Port Numbers]

A) Port 53
B) Port 80
C) Port 25
D) Port 110

✓ **Answer: Port 53**

Q154. What is the default port number used by unencrypted HTTP? [Well-Known Port Numbers]

A) Port 80
B) Port 443
C) Port 8080
D) Port 21

✓ **Answer: Port 80**

Q155. What is the default port number used by encrypted HTTPS? [Well-Known Port Numbers]

A) Port 443
B) Port 80
C) Port 22
D) Port 8443

✓ **Answer: Port 443**

Q156. What is the default port used by SSH for secure remote login? [Well-Known Port Numbers]

A) Port 22

B) Port 23

C) Port 21

D) Port 25

✓ **Answer: Port 22**

Q157. What is the default port for Telnet (insecure remote terminal)? [Well-Known Port Numbers]

A) Port 23

B) Port 22

C) Port 25

D) Port 21

✓ **Answer: Port 23**

Q158. What is the default port for FTP Control connection? [Well-Known Port Numbers]

A) Port 21

B) Port 20

C) Port 22

D) Port 80

✓ **Answer: Port 21**

Q159. What is the default port for active FTP Data connection? [Well-Known Port Numbers]

A) Port 20

B) Port 21

C) Port 25

D) Port 110

✓ **Answer: Port 20**

Q160. What is the standard port for SMTP email transmission? [Well-Known Port Numbers]

A) Port 25

B) Port 110

C) Port 143

D) Port 80

✓ **Answer: Port 25**

Q161. Which port is recommended for modern authenticated email submission using SMTP? [Well-Known Port Numbers]

A) Port 587

B) Port 25

C) Port 110

D) Port 22

✓ **Answer: Port 587**

Q162. What is the default port for standard unencrypted POP3 email retrieval? [Well-Known Port Numbers]

A) Port 110

B) Port 995

C) Port 143

D) Port 25

✓ **Answer: Port 110**

Q163. What is the default port for secure POP3 over SSL/TLS? [Well-Known Port Numbers]

A) Port 995

B) Port 110

C) Port 993

D) Port 465

✓ **Answer: Port 995**

Q164. What is the default port for standard unencrypted IMAP email access? [Well-Known Port Numbers]

A) Port 143

B) Port 993

C) Port 110

D) Port 25

✓ **Answer: Port 143**

Q165. What is the default port for secure IMAP over SSL/TLS? [Well-Known Port Numbers]

A) Port 993

B) Port 143

C) Port 995

D) Port 465

✓ **Answer: Port 993**

UNIT IV: NETWORK LAYER PROTOCOLS

Q166. Which UDP port is used by RIP (Routing Information Protocol) to exchange routing tables? [Well-Known Port Numbers]

A) UDP port 520

B) TCP port 179

C) UDP port 69

D) TCP port 520

✓ **Answer: UDP port 520**

Q167. Which TCP port is used by BGP (Border Gateway Protocol) for inter-domain routing? [Well-Known Port Numbers]

A) TCP port 179

B) UDP port 520

C) TCP port 89

D) UDP port 179

✓ **Answer: TCP port 179**

Q168. Which IP protocol number is used directly by OSPF without requiring a TCP or UDP header? [Well-Known Port Numbers]

A) IP protocol 89

B) TCP port 89

C) UDP port 89

D) IP protocol 6

✓ **Answer: IP protocol 89**

Q169. What is the minimum header size of an IPv4 packet? [Header Sizes & Formats]

A) 20 bytes
B) 40 bytes
C) 8 bytes
D) 16 bytes

✓ **Answer: 20 bytes**

Q170. What is the maximum possible header size of an IPv4 packet including options? [Header Sizes & Formats]

A) 60 bytes
B) 40 bytes
C) 20 bytes
D) 128 bytes

✓ **Answer: 60 bytes**

Q171. What is the fixed base header size of an IPv6 packet? [Header Sizes & Formats]

A) 40 bytes
B) 20 bytes
C) 32 bytes
D) 64 bytes

✓ **Answer: 40 bytes**

UNIT V: TRANSPORT LAYER PROTOCOLS

Q172. What is the standard fixed header size of a UDP datagram? [Header Sizes & Formats]

A) 8 bytes
B) 20 bytes
C) 40 bytes
D) 16 bytes

✓ **Answer: 8 bytes**

Q173. What is the minimum header size of a TCP segment without optional headers? [Header Sizes & Formats]

A) 20 bytes
B) 8 bytes
C) 40 bytes
D) 16 bytes

✓ **Answer: 20 bytes**

Q174. What is the maximum header size of a TCP segment including options? [Header Sizes & Formats]

A) 60 bytes
B) 40 bytes
C) 32 bytes
D) 128 bytes

✓ **Answer: 60 bytes**

UNIT IV: NETWORK LAYER PROTOCOLS

Q175. In the OSI 7-layer model, which layer lies immediately below the Network Layer? [Network Layer Core Principles]

A) Data Link Layer (Layer 2)

B) Physical Layer (Layer 1)

C) Transport Layer (Layer 4)

D) Session Layer (Layer 5)

✓ **Answer: Data Link Layer (Layer 2)**

Q176. In the OSI 7-layer model, which layer lies immediately above the Network Layer? [Network Layer Core Principles]

A) Transport Layer (Layer 4)

B) Data Link Layer (Layer 2)

C) Application Layer (Layer 7)

D) Session Layer (Layer 5)

✓ **Answer: Transport Layer (Layer 4)**

Q177. What type of address is an IP address? [Network Layer Core Principles]

A) Logical network address

B) Physical MAC address

C) Port socket address

D) Hardware serial number

✓ **Answer: Logical network address**

Q178. What is the primary difference between routing and forwarding? [Network Layer Core Principles]

A) Routing determines the overall end-to-end path; forwarding moves a packet from an input interface to an output interface

B) Routing is done by switches; forwarding is done by firewalls

C) Routing modifies MAC addresses; forwarding modifies IP addresses

D) Forwarding calculates shortest paths; routing transmits physical voltages

✓ **Answer: Routing determines the overall end-to-end path; forwarding moves a packet from an input interface to an output interface**

Q179. Why is IP called connectionless? [Network Layer Core Principles]

A) Each packet is treated independently with no prior connection setup between sender and receiver

B) It cannot connect to routers

C) It requires copper cables to connect

D) It only operates when computers are turned off

✓ **Answer: Each packet is treated independently with no prior connection setup between sender and receiver**

Q180. What happens to an IP packet if its Time To Live (TTL) or Hop Limit reaches zero? [Network Layer Core Principles]

A) The packet is discarded and an ICMP Time Exceeded message is sent back to the sender

B) The packet is immediately sent to the destination

C) The packet is stored in router permanent storage

D) The packet is converted into an Ethernet frame

✓ **Answer: The packet is discarded and an ICMP Time Exceeded message is sent back to the sender**

Q181. What is the purpose of the TTL (Time to Live) field in IPv4 packets? [Network Layer Core Principles]

A) To prevent packets from circulating endlessly in routing loops
B) To track how many seconds the computer has been on
C) To measure cable length
D) To encrypt packet contents

✓ **Answer: To prevent packets from circulating endlessly in routing loops**

Q182. What is the equivalent of IPv4's TTL (Time To Live) field in the IPv6 header? [Network Layer Core Principles]

A) Hop Limit
B) Payload Length
C) Next Header
D) Traffic Class

✓ **Answer: Hop Limit**

Q183. What is the equivalent of IPv4's 'Protocol' field in the IPv6 header? [Network Layer Core Principles]

A) Next Header
B) Hop Limit
C) Flow Label
D) Version

✓ **Answer: Next Header**

Q184. What is an Autonomous System (AS)? [Routing Protocols & Algorithms]

A) A collection of routers and networks managed under a single administrative authority
B) A self-driving automobile network
C) A router that operates without electrical power
D) A single client computer connected to Wi-Fi

✓ **Answer: A collection of routers and networks managed under a single administrative authority**

Q185. Which type of routing protocol is used within a single Autonomous System? [Routing Protocols & Algorithms]

A) Interior Gateway Protocol (IGP)
B) Exterior Gateway Protocol (EGP)
C) Border Gateway Protocol
D) Inter-domain Protocol

✓ **Answer: Interior Gateway Protocol (IGP)**

Q186. Which of the following routing protocols is an Exterior Gateway Protocol (EGP)? [Routing Protocols & Algorithms]

A) BGP (Border Gateway Protocol)
B) RIP
C) OSPF
D) IS-IS

✓ **Answer: BGP (Border Gateway Protocol)**

Q187. Which of the following routing protocols are examples of Interior Gateway Protocols (IGPs)? [Routing Protocols & Algorithms]

A) RIP and OSPF
B) BGP and DNS
C) HTTP and FTP
D) TCP and UDP

✓ **Answer: RIP and OSPF**

Q188. Why is RIP unsuitable for large networks or the global Internet? [Routing Protocols & Algorithms]

A) Because its maximum hop limit is 15 and it has slow convergence

B) Because it requires 128-bit addresses only

C) Because it does not support IP packets

D) Because it requires optical fiber only

✓ **Answer: Because its maximum hop limit is 15 and it has slow convergence**

Q189. How frequently does RIP traditionally broadcast or send routing updates to neighbors? [Routing Protocols & Algorithms]

A) Every 30 seconds

B) Every 5 minutes

C) Every 1 millisecond

D) Only when router boots up

✓ **Answer: Every 30 seconds**

Q190. What information does a Link State router send in Link State Advertisements (LSAs)? [Routing Protocols & Algorithms]

A) The state and cost of its directly connected links to all routers in the area

B) Its entire routing table to neighboring routers only

C) All user web passwords

D) Only the default gateway IP

✓ **Answer: The state and cost of its directly connected links to all routers in the area**

Q191. Why do Link State routing protocols converge faster than Distance Vector protocols? [Routing Protocols & Algorithms]

A) Each router has full topology knowledge and computes shortest paths independently using Dijkstra

B) They use slower baud rates

C) They send updates only once a day

D) They do not use IP routing tables

✓ **Answer: Each router has full topology knowledge and computes shortest paths independently using Dijkstra**

Q192. What technique is commonly used to prevent routing loops in Distance Vector routing? [Routing Protocols & Algorithms]

A) Split Horizon and Poison Reverse

B) Dijkstra algorithm

C) Binary search

D) Sliding Window

✓ **Answer: Split Horizon and Poison Reverse**

Q193. In IPv4 multicast, what is the first octet range of Class D addresses? [Multicasting Deep-Dive]

A) 224 to 239

B) 128 to 191

C) 192 to 223

D) 240 to 255

✓ **Answer: 224 to 239**

Q194. Which of the following is a valid IPv4 multicast address? [Multicasting Deep-Dive]

A) 224.0.0.1

B) 192.168.1.1

C) 10.0.0.1

D) 255.255.255.255

✓ **Answer: 224.0.0.1**

Q195. What is the IPv4 multicast address 224.0.0.1 used for? [Multicasting Deep-Dive]

A) All systems/hosts on this subnet

B) All routers on this subnet

C) RIP routers

D) OSPF routers

✓ **Answer: All systems/hosts on this subnet**

Q196. What is the IPv4 multicast address 224.0.0.2 used for? [Multicasting Deep-Dive]

A) All routers on this subnet

B) All hosts on the Internet

C) Default gateway only

D) DHCP servers

✓ **Answer: All routers on this subnet**

Q197. What protocol enables multicast listeners to inform local routers that they wish to receive group traffic in IPv6? [Multicasting Deep-Dive]

A) MLD (Multicast Listener Discovery)

B) IGMP

C) ARP

D) ICMPv4

✓ **Answer: MLD (Multicast Listener Discovery)**

Q198. How many 16-bit blocks (hextets) compose an IPv6 address? [IPv6 Addressing Rules]

A) 8 blocks

B) 4 blocks

C) 16 blocks

D) 32 blocks

✓ **Answer: 8 blocks**

Q199. How is the IPv6 address '2001:0db8:0000:0000:0000:0000:1428:57ab' correctly abbreviated? [IPv6 Addressing Rules]

A) 2001:db8::1428:57ab

B) 2001:db8:0::1428:57ab::

C) 2001:db8:1428:57ab

D) 2001::db8::1428:57ab

✓ **Answer: 2001:db8::1428:57ab**

Q200. Why can the double-colon '::' symbol only be used ONCE in an IPv6 address? [IPv6 Addressing Rules]

A) Because using it more than once creates ambiguity about how many zero blocks were omitted

B) Because hexadecimal notation only allows one colon

C) Because routers will crash

D) Because IPv6 addresses cannot exceed 32 bits

✓ **Answer: Because using it more than once creates ambiguity about how many zero blocks were omitted**

Q201. What is the loopback address in IPv6? [IPv6 Addressing Rules]

A) ::1 (or 0:0:0:0:0:0:0:1)

B) 127.0.0.1

C) ::0

D) fe80::1

✓ **Answer: ::1 (or 0:0:0:0:0:0:0:1)**

Q202. What is the unspecified address in IPv6 representing 'any address' or 'no address'? [IPv6 Addressing Rules]

A) :: (all zeros)

B) ::1

C) ff00::

D) fe80::

✓ **Answer: :: (all zeros)**

Q203. What prefix identifies link-local unicast addresses in IPv6? [IPv6 Addressing Rules]

A) fe80::/10

B) ff00::/8

C) 2001::/16

D) fc00::/7

✓ **Answer: fe80::/10**

UNIT V: TRANSPORT LAYER PROTOCOLS

Q204. What is port multiplexing at the Transport Layer? [Ports & Multiplexing]

A) Combining data streams from multiple applications into packets at sender, and demultiplexing to correct applications at receiver

B) Splitting an optical cable into multiple glass fibers

C) Assigning multiple IP addresses to a single network card

D) Converting analog radio signals into digital packets

✓ **Answer: Combining data streams from multiple applications into packets at sender, and demultiplexing to correct applications at receiver**

Q205. What is the range of Well-Known port numbers reserved for standard protocols? [Ports & Multiplexing]

A) 0 to 1023

B) 1024 to 49151

C) 49152 to 65535

D) 1 to 255

✓ **Answer: 0 to 1023**

Q206. What is the range of Registered port numbers? [Ports & Multiplexing]

A) 1024 to 49151

B) 0 to 1023

C) 49152 to 65535

D) 1000 to 2000

✓ **Answer: 1024 to 49151**

Q207. What is the range of Dynamic, Private, or Ephemeral port numbers automatically chosen by client devices? [Ports & Multiplexing]

A) 49152 to 65535

B) 0 to 1023

C) 1024 to 49151

D) 1 to 65536

✓ **Answer: 49152 to 65535**

Q208. What mechanism does TCP use to perform flow control so a fast sender does not overwhelm a slow receiver? [TCP Features & Mechanisms]

A) Sliding Window flow control (based on Receive Window advertised by receiver)

B) Stop-and-wait ARQ only

C) Dropping packets silently

D) Decreasing physical voltage

✓ **Answer: Sliding Window flow control (based on Receive Window advertised by receiver)**

Q209. What does the Sequence Number in a TCP header represent? [TCP Features & Mechanisms]

A) The byte sequence number of the first data byte in the segment

B) The total count of routers along the path

C) The port number of the receiving program

D) The checksum of the IP header

✓ **Answer: The byte sequence number of the first data byte in the segment**

Q210. What does the Acknowledgment Number in a TCP header represent? [TCP Features & Mechanisms]

A) The next byte sequence number that the receiver expects to receive

B) The last byte that was corrupted

C) The port number of the sender

D) The number of packets dropped

✓ **Answer: The next byte sequence number that the receiver expects to receive**

Q211. What are the four phases of TCP congestion control? [TCP Features & Mechanisms]

A) Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery

B) Open, Connect, Transmit, Close

C) SYN, SYN-ACK, ACK, FIN

D) Listen, Send, Receive, Discard

✓ **Answer: Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery**

Q212. How does TCP detect that a packet was lost in transit? [TCP Features & Mechanisms]

A) Retransmission timeout (RTO) expiration or receipt of 3 duplicate ACKs

B) ICMP error message from the client

C) Receiver sends a broadcast frame

D) The router sends an HTTP 404 response

✓ **Answer: Retransmission timeout (RTO) expiration or receipt of 3 duplicate ACKs**

Q213. What is the role of the URG (Urgent) flag in TCP? [TCP Features & Mechanisms]

A) Indicates that the segment contains urgent data pointed to by the Urgent Pointer field

B) Closes the connection immediately

C) Resets the sliding window to zero

D) Forces packet fragmentation

✓ **Answer: Indicates that the segment contains urgent data pointed to by the Urgent Pointer field**

Q214. What is the role of the PSH (Push) flag in TCP? [TCP Features & Mechanisms]

A) Requests the receiver to push buffered data immediately to the receiving application

B) Terminates the TCP connection

C) Encrypts the payload

D) Suspends transmission

✓ **Answer: Requests the receiver to push buffered data immediately to the receiving application**

Q215. How does SCTP multi-streaming prevent Head-of-Line (HoL) blocking found in TCP? [SCTP In-Depth]

A) Loss or delay in one stream does not block data delivery in other independent streams

B) By converting all data into UDP datagrams

C) By eliminating sequence numbers

D) By limiting message length to 8 bytes

✓ **Answer: Loss or delay in one stream does not block data delivery in other independent streams**

Q216. What is the unit of transmission exchanged between SCTP endpoints? [SCTP In-Depth]

A) SCTP Packets containing one or more Chunks

B) Segments

C) Frames

D) Datagrams only

✓ **Answer: SCTP Packets containing one or more Chunks**

Q217. How does SCTP establish an association between endpoints? [SCTP In-Depth]

A) A 4-way handshake using INIT, INIT-ACK, COOKIE-ECHO, and COOKIE-ACK

B) A 3-way handshake using SYN, SYN-ACK, ACK

C) A 2-way ping-pong exchange

D) No handshake (connectionless)

✓ **Answer: A 4-way handshake using INIT, INIT-ACK, COOKIE-ECHO, and COOKIE-ACK**

Q218. Why does SCTP use a 4-way handshake with signed cookie verification? [SCTP In-Depth]

A) To protect servers against SYN-flood denial of service (DoS) attacks

B) To slow down fast clients

C) To generate random port numbers

D) To replace SSL encryption

✓ **Answer: To protect servers against SYN-flood denial of service (DoS) attacks**

UNIT VI: APPLICATION LAYER

Q219. Which HTTP method is typically used by a client to request data/webpages from a server without modifying server state? [HTTP & Web Protocols]

A) GET

B) POST

C) DELETE

D) PUT

✓ **Answer: GET**

Q220. Which HTTP method is typically used to submit form data or upload files to a server? [HTTP & Web Protocols]

A) POST

B) GET

C) HEAD

D) OPTIONS

✓ **Answer: POST**

Q221. What does HTTP status code 200 represent? [HTTP & Web Protocols]

A) OK (request succeeded)

B) Not Found

C) Internal Server Error

D) Unauthorized

✓ **Answer: OK (request succeeded)**

Q222. What does HTTP status code 404 represent? [HTTP & Web Protocols]

A) Not Found (requested resource does not exist)

B) Bad Request

C) Forbidden

D) Moved Permanently

✓ **Answer: Not Found (requested resource does not exist)**

Q223. What does HTTP status code 500 represent? [HTTP & Web Protocols]

A) Internal Server Error

B) Bad Gateway

C) Gateway Timeout

D) Service Unavailable

✓ **Answer: Internal Server Error**

Q224. What does HTTP status code 301 represent? [HTTP & Web Protocols]

A) Moved Permanently (redirect)

B) Found (temporary redirect)

C) Unauthorized

D) Accepted

✓ **Answer: Moved Permanently (redirect)**

Q225. How do modern web applications maintain session state over stateless HTTP? [HTTP & Web Protocols]

A) Using HTTP Cookies and Session tokens

B) By altering IP addresses on each request

C) By establishing continuous TCP connections that never close

D) By encoding state in DNS records

✓ **Answer: Using HTTP Cookies and Session tokens**

Q226. In DNS, what is the role of a Root Name Server? [DNS Deep-Dive]

A) It directs queries to the appropriate Top-Level Domain (TLD) servers (such as .com, .org, .edu)

B) It stores the IP address of every computer on Earth

C) It translates MAC addresses to IP addresses

D) It runs web browsers for users

✓ **Answer: It directs queries to the appropriate Top-Level Domain (TLD) servers (such as .com, .org, .edu)**

Q227. What is an Authoritative DNS server? [DNS Deep-Dive]

A) A server that holds the actual definitive DNS records for a specific domain name

B) A temporary caching server at an ISP

C) A root server that routes all Internet traffic

D) A browser cache on a local laptop

✓ **Answer: A server that holds the actual definitive DNS records for a specific domain name**

Q228. What is the difference between Recursive and Iterative DNS queries? [DNS Deep-Dive]

A) In recursive queries, the client asks the resolver to find the complete answer; in iterative queries, the server returns the best referral it knows

B) Recursive queries use TCP; iterative queries use UDP

C) Recursive queries are unencrypted; iterative queries use TLS

D) There is no functional difference

✓ **Answer: In recursive queries, the client asks the resolver to find the complete answer; in iterative queries, the server returns the best referral it knows**

Q229. Which DNS record is used for Reverse DNS Lookups (translating an IP address back into a domain name)? [DNS Deep-Dive]

A) PTR (Pointer) record

B) A record

C) MX record

D) CNAME record

✓ **Answer: PTR (Pointer) record**

Q230. What format was developed to allow emails to contain multimedia, attachments, and non-ASCII character sets? [Email & File Protocols]

A) MIME (Multipurpose Internet Mail Extensions)

B) ASCII-7

C) HTML5

D) FTP

✓ **Answer: MIME (Multipurpose Internet Mail Extensions)**

Q231. What is the main limitation of traditional POP3 compared to IMAP? [Email & File Protocols]

A) POP3 downloads emails to one device and deletes or decouples them from the server, preventing synchronized multi-device access

B) POP3 cannot download text emails

C) POP3 requires port 80

D) POP3 only works on Linux

✓ **Answer: POP3 downloads emails to one device and deletes or decouples them from the server, preventing synchronized multi-device access**

Q232. Why does FTP use two separate TCP connections (control and data)? [Email & File Protocols]

A) To allow control commands to be issued while a large data transfer is actively taking place

B) Because one connection cannot transfer more than 1 MB

C) Because control connections use UDP

D) To allow two different users to share one password

✓ **Answer: To allow control commands to be issued while a large data transfer is actively taking place**

Q233. What is the difference between Active FTP and Passive FTP? [Email & File Protocols]

A) In active mode, the server initiates the data connection to the client; in passive mode, the client initiates the data connection (firewall-friendly)

B) Active FTP uses UDP; passive FTP uses TCP

C) Active FTP is encrypted; passive FTP is plaintext

D) Active FTP is only for downloads; passive FTP is for uploads

✓ **Answer: In active mode, the server initiates the data connection to the client; in passive mode, the client initiates the data connection (firewall-friendly)**

Q234. Why is Lossy compression acceptable for image and audio streaming but completely unacceptable for executable programs and source code? [Data Compression Deep-Dive]

A) Because human perception tolerates minor sensory imperfections, but a single lost bit in code causes crashes or corruption

B) Because lossy algorithms cannot run on text files

C) Because lossless compression makes files larger

D) Because audio files cannot be compressed losslessly

✓ **Answer: Because human perception tolerates minor sensory imperfections, but a single lost bit in code causes crashes or corruption**

Q235. Which algorithm is famously used in lossless compression utilities such as GZIP, ZIP, and PNG? [Data Compression Deep-Dive]

A) DEFLATE (combination of LZ77 and Huffman coding)

B) Dijkstra algorithm

C) Bellman-Ford algorithm

D) RSA algorithm

✓ **Answer: DEFLATE (combination of LZ77 and Huffman coding)**

Q236. What type of compression is used by the MP3 audio format? [Data Compression Deep-Dive]

A) Lossy compression (perceptual audio coding removing inaudible frequencies)

B) Lossless compression

C) Run-length encoding without loss

D) Delta modulation without data loss

✓ **Answer: Lossy compression (perceptual audio coding removing inaudible frequencies)**

Q237. What type of compression is used by the JPEG image format? [Data Compression Deep-Dive]

A) Lossy compression (based on Discrete Cosine Transform and quantization)

B) Lossless compression

C) Huffman coding only with zero loss

D) Bitwise copy

✓ **Answer: Lossy compression (based on Discrete Cosine Transform and quantization)**

Q238. What type of compression is used by the GIF and PNG graphic formats? [Data Compression Deep-Dive]

A) Lossless compression

B) Lossy compression

C) Psychoacoustic lossy model

D) DCT quantization

✓ **Answer: Lossless compression**

UNIT IV: NETWORK LAYER

Q239. Which of the following is a fundamental service provided by the Network Layer? [Network Layer Services]

A) Packetizing, logical addressing, routing, and packet forwarding

B) Framing and physical bit transmission

C) Process-to-process port demultiplexing

D) Displaying GUI windows on the monitor

✓ **Answer: Packetizing, logical addressing, routing, and packet forwarding**

Q240. What does 'packetizing' mean at the Network Layer? [Network Layer Services]

A) Encapsulating the transport layer payload into an IP datagram at the source and decapsulating it at destination

B) Converting ASCII text into JPEG format

C) Compressing audio files into MP3

D) Splitting an electrical voltage into radio frequencies

✓ **Answer: Encapsulating the transport layer payload into an IP datagram at the source and decapsulating it at destination**

Q241. Can intermediate routers along a packet's path modify the IP payload data? [Network Layer Services]

A) No, routers do not inspect or modify the upper-layer payload data (except for header fields like TTL)

B) Yes, routers compress the payload at every hop

C) Yes, routers translate payloads between languages

D) Yes, routers encrypt the payload with their private key

✓ **Answer: No, routers do not inspect or modify the upper-layer payload data (except for header fields like TTL)**

Q242. What is the difference between Connectionless service and Connection-oriented service at the Network Layer? [Network Layer Services]

A) Connectionless treats every packet independently without path setup; Connection-oriented sets up a virtual circuit before sending packets

B) Connectionless requires telephone calls; Connection-oriented uses Wi-Fi

C) Connectionless operates at Layer 7; Connection-oriented operates at Layer 1

D) There is no difference in modern networks

✓ **Answer: Connectionless treats every packet independently without path setup; Connection-oriented sets up a virtual circuit before sending packets**

Q243. What is the fundamental principle of Packet Switching in computer networks? [Packet Switching]

A) Data is broken into independent chunks called packets that share network transmission links dynamically

B) A physical copper circuit is reserved exclusively for the entire duration of a call

C) Data is transmitted as continuous unpacketized radio waves

D) Computers exchange data by physically mailing storage drives

✓ **Answer: Data is broken into independent chunks called packets that share network transmission links dynamically**

Q244. Which packet switching approach is used by the Internet Protocol (IP)? [Packet Switching]

A) Datagram approach (connectionless packet switching)

B) Virtual Circuit approach

C) Circuit switching

D) Message switching with magnetic tapes

✓ **Answer: Datagram approach (connectionless packet switching)**

Q245. In a Datagram packet-switched network, how are packets belonging to the same message routed? [Packet Switching]

A) Each packet is routed independently and may follow different paths to the destination

B) All packets must strictly follow the exact same physical wire in locked sequence

C) Packets cannot travel unless all packets arrive at the first router

D) Packets are broadcast to every computer on Earth simultaneously

✓ **Answer: Each packet is routed independently and may follow different paths to the destination**

Q246. What is a defining feature of a Virtual Circuit (VC) packet-switched network (e.g. X.25, Frame Relay, ATM)? [Packet Switching]

A) A predetermined logical path is established during setup, and all packets carry a VC identifier and follow that path

B) Packets are routed with no destination address

C) It does not require any routers or switches

D) It can only transmit voice signals

✓ **Answer: A predetermined logical path is established during setup, and all packets carry a VC identifier and follow that path**

Q247. What is the main advantage of Packet Switching over traditional Circuit Switching? [Packet Switching]

A) Higher link utilization and efficiency through statistical multiplexing and dynamic sharing of bandwidth

B) Guaranteed 100% constant reserved bandwidth even when idle

C) Zero possibility of packet delay or loss

D) Complete elimination of routing tables

✓ **Answer: Higher link utilization and efficiency through statistical multiplexing and dynamic sharing of bandwidth**

Q248. What occurs in Store-and-Forward packet switching at intermediate routers? [Packet Switching]

A) The router must receive the entire packet and verify its checksum before transmitting it onto the outbound link

B) The router forwards individual bits before receiving the full packet

C) The router stores packets on floppy disks permanently

D) The router discards all packets after forwarding

✓ **Answer: The router must receive the entire packet and verify its checksum before transmitting it onto the outbound link**

Q249. What are the four primary components that make up total packet delay in a network? [Network Layer Performance]

A) Transmission delay, Propagation delay, Queuing delay, and Processing delay

B) Encoding delay, Audio delay, Video delay, and Display delay

C) Keyboard delay, Mouse delay, CPU delay, and Disk delay

D) Download delay, Upload delay, Billing delay, and Power delay

✓ **Answer: Transmission delay, Propagation delay, Queuing delay, and Processing delay**

Q250. What is Transmission Delay (D_{trans})? [Network Layer Performance]

A) The time required to push/transmit all the packet's bits into the wire (L / R, packet length / transmission rate)

B) The time for a single bit to travel across the physical medium from sender to receiver

C) The time a packet waits in a router buffer queue

D) The time taken by a router CPU to examine the packet header

✓ **Answer: The time required to push/transmit all the packet's bits into the wire (L / R, packet length / transmission rate)**

Q251. What is Propagation Delay (D_{prop})? [Network Layer Performance]

A) The time for a bit to travel through the physical medium from one end of the link to the other (d / s, distance / propagation speed)

B) The time to push all packet bits onto the link

C) The time spent waiting inside a buffer

D) The time to compute the IP header checksum

✓ Answer: The time for a bit to travel through the physical medium from one end of the link to the other (d / s, distance / propagation speed)

Q252. What is Queuing Delay in network layer performance? [Network Layer Performance]

A) The time a packet spends waiting in router buffer queues for transmission onto the outgoing link

B) The time taken to run the Dijkstra algorithm

C) The time to resolve a DNS query

D) The speed of light in optical fiber

✓ Answer: The time a packet spends waiting in router buffer queues for transmission onto the outgoing link

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Q253. What is Processing Delay in a router? [Network Layer Performance]

A) The time required by the router to examine the packet header, check for bit-level errors, and determine the output link

B) The time to heat up the router silicon

C) The time to push the packet into the buffer queue

D) The propagation delay across transatlantic fiber cables

✓ Answer: The time required by the router to examine the packet header, check for bit-level errors, and determine the output link

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Q254. What is Throughput in computer networks? [Network Layer Performance]

A) The actual rate at which data is successfully transferred from source to destination across a communication link (measured in bps)

B) The maximum theoretically possible frequency of an antenna

C) The size of the router buffer memory

D) The total number of dropped packets

✓ Answer: The actual rate at which data is successfully transferred from source to destination across a communication link (measured in bps)

Q255. What causes Packet Loss at the Network Layer? [Network Layer Performance]

A) When arriving packets find the router buffer queue full (buffer overflow caused by network congestion)

B) When a computer runs out of RAM

C) When the user closes the web browser tab

D) When an IP address contains odd numbers

✓ Answer: When arriving packets find the router buffer queue full (buffer overflow caused by network congestion)

Q256. What is Congestion at the Network Layer? [Network Layer Performance]

A) A state where too many packets are present in a part of the subnet, degrading performance and causing queue overflows

B) A physical break in an optical fiber cable

C) A virus infecting user operating systems

D) A collision between two wireless mice

✓ Answer: A state where too many packets are present in a part of the subnet, degrading performance and causing queue overflows

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Q257. What is the Bandwidth-Delay Product (BDP) in network performance? [Network Layer Performance]

A) The product of link bandwidth and round-trip delay, representing the maximum volume of data in flight in the network pipe

B) The total cost of an ISP connection per month

C) The physical weight of router cables

D) The time to reboot a core switch

✓ **Answer: The product of link bandwidth and round-trip delay, representing the maximum volume of data in flight in the network pipe**

Q258. In IPv4 addressing, what are the two main logical components of an address? [IPv4 Addresses]

A) Network ID (Netid) and Host ID (Hostid)

B) Country Code and Area Code

C) MAC address and Port number

D) Hardware ID and Software ID

✓ **Answer: Network ID (Netid) and Host ID (Hostid)**

Q259. In classful addressing, which class uses the first octet range 1 to 126? [IPv4 Addresses]

A) Class A

B) Class B

C) Class C

D) Class D

✓ **Answer: Class A**

Q260. What is the default subnet mask for an IPv4 Class A network? [IPv4 Addresses]

A) 255.0.0.0 (/8)

B) 255.255.0.0 (/16)

C) 255.255.255.0 (/24)

D) 255.255.255.255 (/32)

✓ **Answer: 255.0.0.0 (/8)**

Q261. In classful addressing, which class uses the first octet range 128 to 191? [IPv4 Addresses]

A) Class B

B) Class A

C) Class C

D) Class D

✓ **Answer: Class B**

Q262. What is the default subnet mask for an IPv4 Class B network? [IPv4 Addresses]

A) 255.255.0.0 (/16)

B) 255.0.0.0 (/8)

C) 255.255.255.0 (/24)

D) 255.240.0.0 (/12)

✓ **Answer: 255.255.0.0 (/16)**

Q263. In classful addressing, which class uses the first octet range 192 to 223? [IPv4 Addresses]

A) Class C

B) Class A

C) Class B

D) Class D

✓ **Answer: Class C**

Q264. What is the default subnet mask for an IPv4 Class C network? [IPv4 Addresses]

A) 255.255.255.0 (/24)

B) 255.255.0.0 (/16)

C) 255.0.0.0 (/8)

D) 255.255.255.128 (/25)

✓ **Answer: 255.255.255.0 (/24)**

Q265. What is IPv4 Class E (240.0.0.0 to 255.255.255.255) reserved for? [IPv4 Addresses]

A) Experimental and research purposes

B) Multicast streaming

C) Commercial web hosting

D) Local loopback testing

✓ **Answer: Experimental and research purposes**

Q266. Why was Classless Inter-Domain Routing (CIDR) introduced to replace classful addressing? [IPv4 Addresses]

A) To prevent IPv4 address exhaustion and reduce the explosion of routing table sizes

B) To force all computers to use hexadecimal notation

C) To eliminate the need for routers

D) To speed up optical laser modulation

✓ **Answer: To prevent IPv4 address exhaustion and reduce the explosion of routing table sizes**

Q267. In CIDR notation '192.168.1.0/26', what does the '/26' prefix length indicate? [IPv4 Addresses]

A) The first 26 bits represent the network prefix, leaving 6 bits for host addressing

B) There are 26 available IP addresses in the subnet

C) The network can have up to 26 routers

D) The packet must be transmitted within 26 milliseconds

✓ **Answer: The first 26 bits represent the network prefix, leaving 6 bits for host addressing**

Q268. How many usable host IP addresses are available in a standard '/24' subnet (e.g. 192.168.1.0/24)? [IPv4 Addresses]

A) 254 usable hosts ($2^8 - 2$)

B) 256 usable hosts

C) 128 usable hosts

D) 512 usable hosts

✓ **Answer: 254 usable hosts ($2^8 - 2$)**

Q269. Why are 2 addresses subtracted when calculating the number of usable hosts in an IPv4 subnet ($2^h - 2$)? [IPv4 Addresses]

A) One address is reserved as the Network Address (all host bits 0), and one as the Broadcast Address (all host bits 1)

B) One for the router and one for the firewall

C) One for DNS and one for DHCP

D) One for the gateway and one for the web server

✓ **Answer: One address is reserved as the Network Address (all host bits 0), and one as the Broadcast Address (all host bits 1)**

Q270. Which of the following represents the private IPv4 address blocks defined in RFC 1918? [IPv4 Addresses]

A) 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16

B) 1.0.0.0/8, 2.0.0.0/8, and 3.0.0.0/8

C) 127.0.0.0/8 and 224.0.0.0/4

D) 192.0.0.0/8 and 240.0.0.0/4

✓ **Answer: 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16**

Q271. Can private IPv4 addresses (such as 192.168.1.10) be routed directly across the public global Internet? [IPv4 Addresses]

A) No, private IP addresses are non-routable on the public Internet and require NAT (Network Address Translation)

B) Yes, all IP addresses route globally without distinction

C) Only if the router uses OSPF

D) Only during night hours

✓ **Answer: No, private IP addresses are non-routable on the public Internet and require NAT (Network Address Translation)**

Q272. What is the purpose of NAT (Network Address Translation)? [IPv4 Addresses]

A) To map private local IP addresses to one or more public IP addresses for Internet access

B) To convert IPv4 packets into IPv6 packets

C) To calculate routing metrics using Dijkstra

D) To assign DNS names to MAC addresses

✓ **Answer: To map private local IP addresses to one or more public IP addresses for Internet access**

Q273. What is the special IPv4 address 127.0.0.1 used for? [IPv4 Addresses]

A) Loopback testing (communicating with the local host itself)

B) Default gateway on home routers

C) Global public broadcast

D) Multicast streaming

✓ **Answer: Loopback testing (communicating with the local host itself)**

Q274. What is the special IPv4 address 255.255.255.255 used for? [IPv4 Addresses]

A) Limited broadcast address (broadcasts to all hosts on the local network)

B) Default route to the Internet

C) Loopback testing

D) DNS root server address

✓ **Answer: Limited broadcast address (broadcasts to all hosts on the local network)**

Q275. What is the IPv4 address 0.0.0.0 used for in routing tables? [IPv4 Addresses]

A) The default route (matches any destination address not specifically listed)

B) The loopback address

C) The broadcast address

D) The multicast rendezvous point

✓ **Answer: The default route (matches any destination address not specifically listed)**

Q276. What principle does a router use to select the matching route when multiple prefixes match an incoming destination IP address? [Forwarding of IP Packets]

A) Longest Prefix Match (LPM)

B) Shortest Prefix Match

C) First match in the text file

D) Random selection

✓ **Answer: Longest Prefix Match (LPM)**

Q277. Suppose a packet destination is 192.168.1.130 and the routing table has entries for '192.168.1.0/24' and '192.168.1.128/25'. Which entry is chosen? [Forwarding of IP Packets]

A) 192.168.1.128/25 because it has the longer prefix match (/25 > /24)

B) 192.168.1.0/24 because it comes first numerically

C) Both routes simultaneously

D) The default route 0.0.0.0/0

✓ **Answer: 192.168.1.128/25 because it has the longer prefix match (/25 > /24)**

Q278. What is the difference between Direct Delivery and Indirect Delivery in IP packet forwarding? [Forwarding of IP Packets]

A) Direct delivery occurs when the destination host is on the same physical network as the deliverer; Indirect delivery forwards the packet to a next-hop router

B) Direct delivery uses optical fiber; Indirect delivery uses copper

C) Direct delivery uses UDP; Indirect delivery uses TCP

D) Direct delivery travels through at least 10 routers

✓ **Answer: Direct delivery occurs when the destination host is on the same physical network as the deliverer; Indirect delivery forwards the packet to a next-hop router**

Q279. What structure inside a high-speed router contains the cached information needed to make forwarding decisions on incoming packets? [Forwarding of IP Packets]

A) Forwarding Information Base (FIB) / Forwarding Table

B) DNS Cache

C) ARP Poison Table

D) MIME table

✓ **Answer: Forwarding Information Base (FIB) / Forwarding Table**

Q280. When an intermediate router forwards an IPv4 datagram to the next hop, what modifications does it make to the IP header? [Forwarding of IP Packets]

A) Decrements the TTL field by at least 1 and recomputes the Header Checksum

B) Changes the Source IP address to the router's IP

C) Replaces the upper-layer payload

D) Inverts all binary bits

✓ **Answer: Decrements the TTL field by at least 1 and recomputes the Header Checksum**

Q281. When a router forwards an IP packet over an Ethernet link, how does it encapsulate the packet? [Forwarding of IP Packets]

A) As the payload of a new Data Link Layer (Ethernet) frame with the next-hop MAC address as destination MAC

B) It sends the IP packet directly without any frame header

C) It converts the IP packet into a DNS record

D) It wraps it in an HTTP header

✓ **Answer: As the payload of a new Data Link Layer (Ethernet) frame with the next-hop MAC address as destination MAC**

Q282. What protocol does an IPv4 router use to find the physical MAC address corresponding to the next-hop IP address? [Forwarding of IP Packets]

A) ARP (Address Resolution Protocol)

B) DNS

C) DHCP

D) ICMP

✓ **Answer: ARP (Address Resolution Protocol)**

Q283. At the network layer, which device is responsible for making routing and forwarding decisions between heterogeneous links? [Network Layer Services]

A) Router

B) Repeater

C) Passive Hub

D) Layer 2 Bridge

✓ **Answer: Router**

Q284. What happens to the transport layer segment when it arrives at the network layer of the sending host? [Network Layer Services]

A) It is encapsulated into an IP datagram with source and destination IP addresses added to the header

B) It is converted into an analog radio wave

C) It is decrypted using AES-256

D) It is split into individual ASCII characters

✓ **Answer: It is encapsulated into an IP datagram with source and destination IP addresses added to the header**

Q285. Which of the following is considered an end-to-end layer rather than a hop-by-hop layer? [Network Layer Services]

A) Transport Layer (Layer 4)

B) Data Link Layer (Layer 2)

C) Physical Layer (Layer 1)

D) Network Layer hop routing

✓ **Answer: Transport Layer (Layer 4)**

Q286. What is the term for the maximum amount of data that a link can transmit in a given time period? [Packet Switching]

A) Bandwidth (or link capacity)

B) Propagation delay

C) Attenuation factor

D) Hop metric

✓ **Answer: Bandwidth (or link capacity)**

Q287. In packet-switched networks, what problem occurs when packets of the same message take different paths with varying delays? [Packet Switching]

A) Out-of-order packet arrival at the destination

B) Complete loss of electricity on the cable

C) Physical destruction of routers

D) Permanent corruption of hard disk drives

✓ **Answer: Out-of-order packet arrival at the destination**

Q288. Which layer is primarily responsible for reordering out-of-order datagrams so the application receives an in-order byte stream? [Packet Switching]

A) Transport Layer (via TCP sequence numbers)

B) Network Layer (IP)

C) Physical Layer

D) Data Link Layer

✓ **Answer: Transport Layer (via TCP sequence numbers)**

Q289. What header field in an IPv4 packet ensures that packets do not loop endlessly during temporary routing loops? [Packet Switching]

A) TTL (Time To Live)

B) Checksum

C) Identification

D) TOS (Type of Service)

✓ **Answer: TTL (Time To Live)**

Q290. What is the typical default initial TTL value set by many modern operating systems in IP datagrams? [Packet Switching]

A) 64 or 128

B) 1

C) 16

D) 1024

✓ **Answer: 64 or 128**

Q291. In a Virtual Circuit network, what table do switches maintain to map incoming VC identifiers to outgoing VC identifiers and ports? [Packet Switching]

A) VC Translation Table (or Label Information Base)

B) ARP Table

C) DNS Cache

D) DHCP Lease Table

✓ **Answer: VC Translation Table (or Label Information Base)**

Q292. If a packet has a length of $L = 10,000$ bits and the transmission link rate is $R = 10$ Mbps ($10,000,000$ bps), what is the Transmission Delay (D_{trans})? [Network Layer Performance]

A) 1 millisecond (0.001 s)

B) 10 milliseconds

C) 0.1 millisecond

D) 1 second

✓ **Answer: 1 millisecond (0.001 s)**

Q293. If a fiber cable length is $d = 2,000$ km and light travels in fiber at speed $s = 200,000$ km/s, what is the Propagation Delay (D_{prop})? [Network Layer Performance]

A) 10 milliseconds (0.01 s)

B) 1 millisecond

C) 100 milliseconds

D) 0.1 second

✓ **Answer: 10 milliseconds (0.01 s)**

Q294. Does the Transmission Delay (L/R) depend on the physical distance between the sender and receiver? [Network Layer Performance]

A) No, it depends only on the packet length and link bandwidth rate

B) Yes, transmission delay increases linearly with fiber distance

C) Yes, transmission delay doubles every 100 km

D) Only when using satellite links

✓ **Answer: No, it depends only on the packet length and link bandwidth rate**

Q295. Does the Propagation Delay (d/s) depend on the packet size in bits? [Network Layer Performance]

A) No, it depends only on the physical distance and transmission medium propagation speed

B) Yes, larger packets propagate slower through glass fiber

C) Yes, 1500-byte packets travel at half the speed of light

D) Only when routers have full buffers

✓ **Answer: No, it depends only on the physical distance and transmission medium propagation speed**

Q296. What term describes the variation in packet arrival delay across a network connection? [Network Layer Performance]

A) Jitter

B) Attenuation

C) Crosstalk

D) Dispersion

✓ **Answer: Jitter**

Q297. Why is low jitter critically important for real-time applications such as VoIP and live video conferencing? [Network Layer Performance]

A) High jitter causes audio gaps, jerky video playback, and buffer underflows

B) High jitter reduces operating system battery life

C) High jitter causes physical damage to routers

D) High jitter changes the font of web pages

✓ **Answer: High jitter causes audio gaps, jerky video playback, and buffer underflows**

Q298. What is Round Trip Time (RTT)? [Network Layer Performance]

A) The time taken for a packet to travel from source to destination plus the time for its acknowledgment to return

B) The time to boot up a router switch

C) The time taken to generate an IP address

D) The propagation delay in copper wires only

✓ **Answer: The time taken for a packet to travel from source to destination plus the time for its acknowledgment to return**

Q299. What is the total theoretical address space of IPv4? [IPv4 Addresses]

A) $2^{32} \approx 4.29$ billion addresses

B) 2^{64} addresses

C) $2^{16} = 65,536$ addresses

D) 2^{128} addresses

✓ **Answer: $2^{32} \approx 4.29$ billion addresses**

Q300. What are the leading bits of an IPv4 Class A address in binary? [IPv4 Addresses]

A) 0

B) 10

C) 110

D) 1110

✓ **Answer: 0**

Q301. What are the leading bits of an IPv4 Class B address in binary? [IPv4 Addresses]

A) 10

B) 0

C) 110

D) 1110

✓ **Answer: 10**

Q302. What are the leading bits of an IPv4 Class C address in binary? [IPv4 Addresses]

A) 110

B) 10

C) 0

D) 1110

✓ **Answer: 110**

Q303. What are the leading bits of an IPv4 Class D (multicast) address in binary? [IPv4 Addresses]

A) 1110

B) 110

C) 10

D) 1111

✓ **Answer: 1110**

Q304. What is the maximum number of usable hosts in an IPv4 Class C network? [IPv4 Addresses]

A) 254 hosts

B) 65,534 hosts

C) 16,777,214 hosts

D) 128 hosts

✓ **Answer: 254 hosts**

Q305. What is the maximum number of usable hosts in an IPv4 Class B network? [IPv4 Addresses]

A) 65,534 hosts ($2^{16} - 2$)

B) 254 hosts

C) 1,024 hosts

D) 16,777,214 hosts

✓ **Answer: 65,534 hosts ($2^{16} - 2$)**

Q306. What is the maximum number of usable hosts in an IPv4 Class A network? [IPv4 Addresses]

A) 16,777,214 hosts ($2^{24} - 2$)

B) 65,534 hosts

C) 254 hosts

D) 4,294,967,296 hosts

✓ **Answer: 16,777,214 hosts ($2^{24} - 2$)**

Q307. What is the subnet mask representation in dotted decimal for a '/25' prefix? [IPv4 Addresses]

A) 255.255.255.128

B) 255.255.255.192

C) 255.255.255.224

D) 255.255.255.0

✓ **Answer: 255.255.255.128**

Q308. What is the subnet mask representation in dotted decimal for a '/26' prefix? [IPv4 Addresses]

A) 255.255.255.192

B) 255.255.255.128

C) 255.255.255.224

D) 255.255.255.240

✓ **Answer: 255.255.255.192**

Q309. What is the subnet mask representation in dotted decimal for a '/27' prefix? [IPv4 Addresses]

A) 255.255.255.224

B) 255.255.255.240

C) 255.255.255.192

D) 255.255.255.128

✓ **Answer: 255.255.255.224**

Q310. What is the subnet mask representation in dotted decimal for a '/28' prefix? [IPv4 Addresses]

A) 255.255.255.240

B) 255.255.255.224

C) 255.255.255.248

D) 255.255.255.252

✓ **Answer: 255.255.255.240**

Q311. What is the subnet mask representation in dotted decimal for a '/30' prefix (commonly used for point-to-point router links)? [IPv4 Addresses]

A) 255.255.255.252

B) 255.255.255.248

C) 255.255.255.254

D) 255.255.255.240

✓ **Answer: 255.255.255.252**

Q312. How many usable host IP addresses are provided by a '/30' subnet? [IPv4 Addresses]

A) 2 usable host addresses ($2^2 - 2$)

B) 4 usable hosts

C) 6 usable hosts

D) 1 usable host

✓ **Answer: 2 usable host addresses ($2^2 - 2$)**

Q313. In the IP address 192.168.10.65/26, what is the Network Address of the subnet? [IPv4 Addresses]

A) 192.168.10.64

B) 192.168.10.0

C) 192.168.10.128

D) 192.168.10.65

✓ **Answer: 192.168.10.64**

Q314. In the IP address 192.168.10.65/26, what is the Directed Broadcast Address of the subnet? [IPv4 Addresses]

A) 192.168.10.127

B) 192.168.10.255

C) 192.168.10.64

D) 192.168.10.128

✓ **Answer: 192.168.10.127**

Q315. What is the network address of the private class A address block defined in RFC 1918? [IPv4 Addresses]

A) 10.0.0.0/8 (10.0.0.0 to 10.255.255.255)

B) 172.16.0.0/12

C) 192.168.0.0/16

D) 11.0.0.0/8

✓ **Answer: 10.0.0.0/8 (10.0.0.0 to 10.255.255.255)**

Q316. What is the address range of the private class B address block defined in RFC 1918? [IPv4 Addresses]

A) 172.16.0.0 to 172.31.255.255 (/12)

B) 172.0.0.0 to 172.15.255.255

C) 172.16.0.0 to 172.16.255.255

D) 172.32.0.0 to 172.64.255.255

✓ **Answer: 172.16.0.0 to 172.31.255.255 (/12)**

Q317. What is the address range of the private class C address block defined in RFC 1918? [IPv4 Addresses]

A) 192.168.0.0 to 192.168.255.255 (/16)

B) 192.0.0.0 to 192.255.255.255

C) 192.168.1.0 to 192.168.1.255 only

D) 192.170.0.0 to 192.170.255.255

✓ **Answer: 192.168.0.0 to 192.168.255.255 (/16)**

Q318. What is an APIPA (Automatic Private IP Addressing) address assigned by Windows when DHCP fails? [IPv4 Addresses]

A) 169.254.0.0/16 (169.254.0.1 to 169.254.255.254)

B) 192.168.1.1

C) 127.0.0.1

D) 10.0.0.1

✓ **Answer: 169.254.0.0/16 (169.254.0.1 to 169.254.255.254)**

Q319. What is the size of the 'Version' field in an IPv4 header? [IPv4 Header Structure]

A) 4 bits (value = 4)

B) 8 bits

C) 16 bits

D) 32 bits

✓ **Answer: 4 bits (value = 4)**

Q320. What does the Header Length (IHL) field in an IPv4 header specify? [IPv4 Header Structure]

A) The total length of the IP header in 4-byte (32-bit) words

B) The length of the payload in bytes

C) The number of routers along the path

D) The MTU of the outgoing interface

✓ **Answer: The total length of the IP header in 4-byte (32-bit) words**

Q321. If the IHL (Internet Header Length) field in an IPv4 header has a decimal value of 5, what is the actual header length in bytes? [IPv4 Header Structure]

A) 20 bytes (5×4)

B) 5 bytes

C) 10 bytes

D) 40 bytes

✓ **Answer: 20 bytes (5×4)**

Q322. What is the maximum value of the Total Length field in an IPv4 header? [IPv4 Header Structure]

A) 65,535 bytes (16 bits)

B) 20 bytes

C) 1,500 bytes

D) 4,294,967,295 bytes

✓ **Answer: 65,535 bytes (16 bits)**

Q323. What are the three flags in the IPv4 header used for fragmentation control? [IPv4 Header Structure]

A) Reserved (0), Don't Fragment (DF), and More Fragments (MF)

B) SYN, ACK, FIN

C) URG, PSH, RST

D) TOS, TTL, IHL

✓ **Answer: Reserved (0), Don't Fragment (DF), and More Fragments (MF)**

Q324. What happens if an IPv4 packet larger than the link MTU arrives at a router and its 'DF' (Don't Fragment) flag is set to 1? [IPv4 Header Structure]

A) The packet is dropped and the router returns an ICMP 'Fragmentation Needed and DF set' error

B) The router forces fragmentation anyway

C) The router compresses the packet with ZIP

D) The packet is converted into a broadcast frame

✓ **Answer: The packet is dropped and the router returns an ICMP 'Fragmentation Needed and DF set' error**

Q325. What unit is used for the Fragment Offset field in the IPv4 header? [IPv4 Header Structure]

A) 8-byte (64-bit) blocks

B) Single bytes

C) 32-bit words

D) 16-byte blocks

✓ **Answer: 8-byte (64-bit) blocks**

Q326. What is the Protocol field value in the IPv4 header for TCP? [IPv4 Header Structure]

A) 6

B) 17

C) 1

D) 89

✓ **Answer: 6**

Q327. What is the Protocol field value in the IPv4 header for UDP? [IPv4 Header Structure]

A) 17

B) 6

C) 1

D) 89

✓ **Answer: 17**

Q328. What is the Protocol field value in the IPv4 header for ICMP? [IPv4 Header Structure]

A) 1

B) 6

C) 17

D) 2

✓ **Answer: 1**

Q329. What is the Protocol field value in the IPv4 header for IGMP? [IPv4 Header Structure]

A) 2

B) 1

C) 6

D) 17

✓ **Answer: 2**

Q330. What does the IPv4 Header Checksum protect against? [IPv4 Header Structure]

A) Bit errors in the IP header only (it does NOT cover the payload)

B) Errors in the entire packet payload

C) Malicious cyber attacks

D) Router hardware failure

✓ **Answer: Bit errors in the IP header only (it does NOT cover the payload)**

Q331. What does a routing table entry with next-hop 'Direct' or '0.0.0.0' indicate? [Forwarding of IP Packets]

A) The destination host resides on the locally connected network attached to that interface

B) The packet should be discarded

C) The packet should be sent to the ISP

D) The packet requires satellite uplink

✓ **Answer: The destination host resides on the locally connected network attached to that interface**

Q332. What is the purpose of the Default Gateway on a host computer? [Forwarding of IP Packets]

A) It is the local router IP address to which the host forwards packets destined for remote subnets

B) It is the root DNS server address

C) It is the public IP of Google

D) It is the MAC address of the host network card

✓ **Answer: It is the local router IP address to which the host forwards packets destined for remote subnets**

Q333. What algorithm data structure is commonly used inside routers to perform fast Longest Prefix Match (LPM) lookups? [Forwarding of IP Packets]

A) Trie (Prefix Tree) / Patricia Trie

B) Linked List

C) Bubble Sort

D) FIFO Stack

✓ **Answer: Trie (Prefix Tree) / Patricia Trie**

Q334. What is the metric used by OSPF to compute path cost? [Routing Protocols & Algorithms]

A) Cost based on reference bandwidth ($10^8 / \text{bandwidth in bps}$)

B) Hop count only

C) Packet size

D) Physical fiber color

✓ **Answer: Cost based on reference bandwidth ($10^8 / \text{bandwidth in bps}$)**

Q335. What is Area 0 in an OSPF multi-area network? [Routing Protocols & Algorithms]

A) The Backbone Area (to which all other OSPF areas must connect)

B) The isolated private stub area

C) The default route area

D) The external internet gateway

✓ **Answer: The Backbone Area (to which all other OSPF areas must connect)**

Q336. In OSPF, what is an ABR (Area Border Router)? [Routing Protocols & Algorithms]

A) A router that connects one or more non-backbone areas to the backbone area (Area 0)

B) A router connecting to a competitor ISP

C) A home Wi-Fi router

D) A core Ethernet switch

✓ **Answer: A router that connects one or more non-backbone areas to the backbone area (Area 0)**

Q337. In OSPF, what is an ASBR (Autonomous System Boundary Router)? [Routing Protocols & Algorithms]

A) A router that exchanges routing information with routers belonging to other Autonomous Systems

B) A bridge between Layer 1 and Layer 2

C) A local DHCP server

D) An access point for smartphones

✓ **Answer: A router that exchanges routing information with routers belonging to other Autonomous Systems**

Q338. In IPv6, how are optional headers handled compared to IPv4? [IPv6 Protocol Deep-Dive]

A) Using Extension Headers chained together sequentially via the 'Next Header' field

B) Appended as variable fields inside the base 40-byte header

C) Included inside the TCP payload

D) Sent as separate ICMP packets

✓ **Answer: Using Extension Headers chained together sequentially via the 'Next Header' field**

Q339. What is the Flow Label field (20 bits) in the IPv6 base header used for? [IPv6 Protocol Deep-Dive]

A) To identify packets belonging to the same flow for Quality of Service (QoS) handling without opening extension headers

B) To measure packet length

C) To specify the encryption algorithm

D) To count the number of routers

✓ **Answer: To identify packets belonging to the same flow for Quality of Service (QoS) handling without opening extension headers**

Q340. What is the Traffic Class field (8 bits) in the IPv6 header equivalent to in IPv4? [IPv6 Protocol Deep-Dive]

A) Type of Service (TOS) / Differentiated Services (DiffServ)

B) Time To Live (TTL)

C) Identification

D) Fragment Offset

✓ **Answer: Type of Service (TOS) / Differentiated Services (DiffServ)**

Q341. What mechanism does IPv6 use to discover the Maximum Transmission Unit (MTU) of the path before sending data? [IPv6 Protocol Deep-Dive]

A) Path MTU Discovery (PMTUD) using ICMPv6 Packet Too Big messages

B) ARP broadcast probing

C) Trial and error by rebooting routers

D) Fixed 576-byte fallback only

✓ **Answer: Path MTU Discovery (PMTUD) using ICMPv6 Packet Too Big messages**

Q342. What is the minimum MTU that every IPv6 link must support? [IPv6 Protocol Deep-Dive]

A) 1,280 bytes

B) 576 bytes

C) 1,500 bytes

D) 65,535 bytes

✓ **Answer: 1,280 bytes**

Q343. What is SLAAC in IPv6 networking? [IPv6 Protocol Deep-Dive]

A) Stateless Address Autoconfiguration (allows hosts to automatically configure their own IPv6 address using Router Advertisements)

B) Standard Link Addressing and Authorization Code

C) Synchronous Line Access and Allocation Channel

D) Software Layer Automated Access Control

✓ **Answer: Stateless Address Autoconfiguration (allows hosts to automatically configure their own IPv6 address using Router Advertisements)**

UNIT V: TRANSPORT LAYER

Q344. What is the port number size in bits for both TCP and UDP? [Transport Layer Services]

A) 16 bits (0 to 65,535)

B) 32 bits

C) 8 bits

D) 64 bits

✓ **Answer: 16 bits (0 to 65,535)**

Q345. What is a Socket in network programming? [Transport Layer Services]

A) An endpoint of a two-way communication link, uniquely defined by an IP address and Port number tuple

B) A physical electrical wall outlet

C) A software virus scanner

D) A cable connector for fiber optics

✓ **Answer: An endpoint of a two-way communication link, uniquely defined by an IP address and Port number tuple**

Q346. What is a 4-tuple that uniquely identifies a TCP connection on the Internet? [Transport Layer Services]

A) (Source IP, Source Port, Destination IP, Destination Port)

B) (MAC Source, MAC Destination, IP Source, IP Destination)

C) (URL, Username, Password, Token)

D) (Browser, Operating System, Screen Size, RAM)

✓ **Answer: (Source IP, Source Port, Destination IP, Destination Port)**

Q347. In UDP, what does the Length field in the header represent? [User Datagram Protocol (UDP)]

A) The total length of the UDP segment in bytes (Header + Data)

B) The length of the data payload only

C) The MTU of the network card

D) The count of dropped packets

✓ **Answer: The total length of the UDP segment in bytes (Header + Data)**

Q348. What is the minimum possible value of the UDP Length field? [User Datagram Protocol (UDP)]

A) 8 bytes (when the UDP payload data length is 0)

B) 0 bytes

C) 20 bytes

D) 1 byte

✓ **Answer: 8 bytes (when the UDP payload data length is 0)**

Q349. Is the UDP checksum calculation mandatory or optional in IPv4? [User Datagram Protocol (UDP)]

A) Optional in IPv4, but mandatory in IPv6

B) Mandatory in IPv4, optional in IPv6

C) Optional in both

D) Mandatory in both

✓ **Answer: Optional in IPv4, but mandatory in IPv6**

Q350. What pseudo-header is included during the calculation of the UDP and TCP checksums? [User Datagram Protocol (UDP)]

A) A pseudo-header containing Source IP, Destination IP, Protocol, and Length from the IP layer

B) The MAC address header

C) The HTTP request line

D) The physical layer preamble

✓ **Answer: A pseudo-header containing Source IP, Destination IP, Protocol, and Length from the IP layer**

Q351. In TCP, what is Cumulative Acknowledgment? [TCP Protocol Details]

A) The ACK number acknowledges all bytes received up to that number minus 1

B) An ACK is sent for every single individual bit

C) An ACK confirms only the last byte received, ignoring prior gaps

D) Acknowledgments accumulate until the computer is rebooted

✓ Answer: The ACK number acknowledges all bytes received up to that number minus 1

Q352. What is the TCP Maximum Segment Size (MSS)? [TCP Protocol Details]

A) The maximum amount of application data (excluding TCP/IP headers) that a host can receive in a single TCP segment

B) The maximum size of the Ethernet cable

C) The total size of the TCP header with options

D) The maximum number of TCP sockets permitted on a computer

✓ Answer: The maximum amount of application data (excluding TCP/IP headers) that a host can receive in a single TCP segment

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Q353. How is the TCP MSS (Maximum Segment Size) calculated for standard Ethernet (MTU = 1500 bytes)? [TCP Protocol Details]

A) $1500 - 20 \text{ (IP header)} - 20 \text{ (TCP header)} = 1460 \text{ bytes}$

B) $1500 - 8 = 1492$ bytes

C) $1500 - 40 = 1460$ bytes

D) $1500 \times 2 = 3000$ bytes

✓ Answer: $1500 - 20 \text{ (IP header)} - 20 \text{ (TCP header)} = 1460 \text{ bytes}$

Q354. What is the role of the TCP Receive Window (rwnd) field? [TCP Protocol Details]

A) To communicate available buffer space in the receiver so the sender does not overflow the buffer (Flow Control)

B) To indicate the speed of the sender's CPU

C) To select an encryption key

D) To measure packet round-trip delay

✓ Answer: To communicate available buffer space in the receiver so the sender does not overflow the buffer (Flow Control)

Q355. What is the Congestion Window (cwnd) in TCP? [TCP Protocol Details]

A) A state variable maintained by the sender that limits transmission to avoid congesting the network path

B) A physical window on the router casing

C) The buffer size advertised by the receiver

D) The size of the hard drive cache

✓ Answer: A state variable maintained by the sender that limits transmission to avoid congesting the network path

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Q356. In TCP, what limits the maximum number of unacknowledged bytes a sender can transmit at any moment? [TCP Protocol Details]

A) $\min(\text{rwnd}, \text{cwnd})$ (the minimum of receiver window and congestion window)

B) $\text{rwnd} + \text{cwnd}$

C) Only the speed of light

D) The total RAM on the server

✓ Answer: $\min(\text{rwnd}, \text{cwnd})$ (the minimum of receiver window and congestion window)

Q357. During the TCP Slow Start phase, how does the Congestion Window (cwnd) increase upon receiving ACKs? [TCP Protocol Details]

A) Exponentially (doubles every RTT until threshold ssthresh is reached)

B) Linearly by 1 segment per RTT

C) It decreases by half

D) It remains constant

✓ **Answer: Exponentially (doubles every RTT until threshold ssthresh is reached)**

Q358. During the TCP Congestion Avoidance phase, how does the Congestion Window (cwnd) increase? [TCP Protocol Details]

A) Linearly (increases by 1 MSS per RTT)

B) Exponentially (doubles every RTT)

C) Quadratically

D) Logarithmically

✓ **Answer: Linearly (increases by 1 MSS per RTT)**

Q359. What action triggers Fast Retransmit in TCP? [TCP Protocol Details]

A) Arrival of 3 duplicate ACKs (indicating a missing segment while subsequent segments arrived)

B) Expiration of the Retransmission Timer (RTO)

C) A user clicking refresh in a browser

D) A packet checksum failure in the router

✓ **Answer: Arrival of 3 duplicate ACKs (indicating a missing segment while subsequent segments arrived)**

Q360. In TCP connection teardown, how many steps (or packets) are typically exchanged? [TCP Protocol Details]

A) 4-way termination (FIN -> ACK, FIN -> ACK)

B) 3-way handshake

C) 2-way reset

D) 1-way abort

✓ **Answer: 4-way termination (FIN -> ACK, FIN -> ACK)**

Q361. What is the purpose of the TIME_WAIT state in the TCP state machine on the endpoint that initiated active close? [TCP Protocol Details]

A) To ensure the final ACK was received by the remote peer and allow late-arriving duplicate segments to expire from the network

B) To download pending web images

C) To recharge the network interface card

D) To wait for a user password prompt

✓ **Answer: To ensure the final ACK was received by the remote peer and allow late-arriving duplicate segments to expire from the network**

Q362. How long does a TCP connection typically remain in the TIME_WAIT state? [TCP Protocol Details]

A) $2 \times \text{MSL}$ (Maximum Segment Lifetime, typically 1 to 2 minutes)

B) 5 seconds

C) 24 hours

D) 10 milliseconds

✓ **Answer: $2 \times \text{MSL}$ (Maximum Segment Lifetime, typically 1 to 2 minutes)**

Q363. In SCTP, what are the two main categories of Chunks found inside an SCTP packet? [SCTP Deep-Dive]

A) Control Chunks and Data Chunks

B) Header Chunks and Trailer Chunks

C) TCP Chunks and UDP Chunks

D) Binary Chunks and ASCII Chunks

✓ **Answer: Control Chunks and Data Chunks**

Q364. Can an SCTP packet contain multiple chunks bundled together in a single transmission? [SCTP Deep-Dive]

A) Yes, an SCTP packet can bundle control and data chunks up to the path MTU

B) No, SCTP allows strictly one chunk per packet

C) Only if all chunks are encrypted

D) Only during association shutdown

✓ **Answer: Yes, an SCTP packet can bundle control and data chunks up to the path MTU**

Q365. What does SCTP use to acknowledge received data chunks? [SCTP Deep-Dive]

A) SACK (Selective Acknowledgment) chunk

B) ACK flag bit

C) SYN packet

D) ICMP Echo Reply

✓ **Answer: SACK (Selective Acknowledgment) chunk**

Q366. What is Transmission Sequence Number (TSN) in SCTP? [SCTP Deep-Dive]

A) A sequence number assigned to each data chunk for end-to-end reliable delivery and fragmentation reassembly

B) The port number of the sender

C) The IP address of the gateway

D) The stream identifier within the application

✓ **Answer: A sequence number assigned to each data chunk for end-to-end reliable delivery and fragmentation reassembly**

Q367. What is Stream Identifier (SI) and Stream Sequence Number (SSN) in SCTP? [SCTP Deep-Dive]

A) SI identifies the specific stream, and SSN preserves the order of messages within that individual stream

B) They replace the MAC address

C) They indicate the size of the packet

D) They specify the time of day

✓ **Answer: SI identifies the specific stream, and SSN preserves the order of messages within that individual stream**

UNIT VI: APPLICATION LAYER

Q368. What is the difference between Persistent and Non-Persistent HTTP connections? [HTTP & Web Details]

A) Persistent HTTP allows multiple requests and responses over a single TCP connection; Non-persistent closes the TCP connection after every single object

B) Persistent HTTP uses UDP; Non-persistent uses TCP

C) Persistent HTTP works only without images

D) Persistent HTTP requires user credit cards

✓ **Answer: Persistent HTTP allows multiple requests and responses over a single TCP connection; Non-persistent closes the TCP connection after every single object**

Q369. Which version of HTTP introduced persistent connections by default? [HTTP & Web Details]

A) HTTP/1.1

B) HTTP/0.9

C) HTTP/1.0

D) Telnet 1.0

✓ **Answer: HTTP/1.1**

Q370. What major feature did HTTP/2 introduce to solve Head-of-Line blocking at the application layer? [HTTP & Web Details]

A) Binary framing and multiplexing of multiple streams over a single TCP connection

B) Switching to pure UDP

C) Removing all header text

D) Eliminating server status codes

✓ **Answer: Binary framing and multiplexing of multiple streams over a single TCP connection**

Q371. What underlying transport protocol does HTTP/3 use instead of TCP? [HTTP & Web Details]

A) QUIC (Quick UDP Internet Connections over UDP)

B) Raw IP datagrams

C) SCTP

D) Ethernet frames

✓ **Answer: QUIC (Quick UDP Internet Connections over UDP)**

Q372. What is an HTTP Request Header 'User-Agent' used for? [HTTP & Web Details]

A) It identifies the client application, browser type, rendering engine, and operating system making the request

B) It contains the user's credit card PIN

C) It specifies the IP address of the router

D) It generates random passwords

✓ **Answer: It identifies the client application, browser type, rendering engine, and operating system making the request**

Q373. What is the purpose of the HTTP 'Cache-Control' header? [HTTP & Web Details]

A) Directs browsers and intermediate proxies whether and for how long to store copies of web resources locally

B) Controls the speed of the CPU

C) Resets the router firewall

D) Encrypts the database

✓ **Answer: Directs browsers and intermediate proxies whether and for how long to store copies of web resources locally**

Q374. What does HTTP status code 403 represent? [HTTP & Web Details]

A) Forbidden (the server understands the request but refuses to authorize it)

B) Not Found

C) Bad Gateway

D) OK

✓ **Answer: Forbidden (the server understands the request but refuses to authorize it)**

Q375. What does HTTP status code 503 represent? [HTTP & Web Details]

A) Service Unavailable (server is temporarily overloaded or down for maintenance)

B) Not Found

C) Unauthorized

D) Moved Temporarily

✓ **Answer: Service Unavailable (server is temporarily overloaded or down for maintenance)**

Q376. What does HTTP status code 304 represent? [HTTP & Web Details]

A) Not Modified (resource has not changed since the version specified in client cache)

B) Internal Server Error

C) Created

D) Accepted

✓ **Answer: Not Modified (resource has not changed since the version specified in client cache)**

Q377. What is the maximum length of a domain name in the DNS hierarchy? [DNS Protocol Details]

A) 253 characters (labels separated by dots, max 63 characters per label)

B) 1,024 characters

C) 32 characters

D) 64 characters

✓ **Answer: 253 characters (labels separated by dots, max 63 characters per label)**

Q378. What is a DNS Resolver? [DNS Protocol Details]

A) A client program or local ISP server that translates domain names into IP addresses by querying DNS servers

B) A hardware cable connector

C) A program that formats hard drives

D) A software compiler for C++

✓ **Answer: A client program or local ISP server that translates domain names into IP addresses by querying DNS servers**

Q379. What is DNS Caching, and why is it used? [DNS Protocol Details]

A) Storing previously resolved domain lookups locally for a period defined by TTL to reduce lookup latency and network traffic

B) Deleting old domain names from the Internet

C) Encrypting passwords on the local hard disk

D) Compressing web page images

✓ **Answer: Storing previously resolved domain lookups locally for a period defined by TTL to reduce lookup latency and network traffic**

Q380. What does the TTL (Time to Live) field in a DNS Resource Record indicate? [DNS Protocol Details]

A) The number of seconds a caching resolver is permitted to keep the record in cache before discarding it

B) The number of routers the query can cross

C) The bandwidth of the DNS server

D) The age of the domain owner

✓ **Answer: The number of seconds a caching resolver is permitted to keep the record in cache before discarding it**

Q381. What is a DNS Zone Transfer, and which transport protocol does it use? [DNS Protocol Details]

A) Replicating DNS database records from a primary master DNS server to secondary slave servers using TCP port 53

B) Sending an email using UDP

C) Uploading files via FTP

D) Resolving an IP using ICMP

✓ **Answer: Replicating DNS database records from a primary master DNS server to secondary slave servers using TCP port 53**

Q382. Which DNS record is used to store human-readable arbitrary text, often used for SPF email verification and domain ownership validation? [DNS Protocol Details]

A) TXT record

B) A record

C) PTR record

D) NS record

✓ **Answer: TXT record**

Q383. Which DNS record defines the start of authority for a zone, including primary master name server, administrator email, and serial number? [DNS Protocol Details]

A) SOA (Start of Authority) record

B) MX record

C) A record

D) CNAME record

✓ **Answer: SOA (Start of Authority) record**

Q384. What protocol extension allows SMTP to secure communication with TLS encryption over port 587 or 25? [Email Protocols Details]

A) STARTTLS

B) SMTPSSL

C) SECUREMAIL

D) OPENSSL

✓ **Answer: STARTTLS**

Q385. What are the two major components of an email system architecture? [Email Protocols Details]

A) User Agent (MUA) and Mail Transfer Agent (MTA)

B) Web Browser and Web Server

C) Client and Router

D) Encoder and Decoder

✓ **Answer: User Agent (MUA) and Mail Transfer Agent (MTA)**

Q386. Which of the following software programs is an example of a Mail User Agent (MUA)? [Email Protocols Details]

A) Microsoft Outlook / Mozilla Thunderbird

B) Postfix

C) BIND

D) Apache HTTP Server

✓ **Answer: Microsoft Outlook / Mozilla Thunderbird**

Q387. Which of the following software programs is an example of a Mail Transfer Agent (MTA)? [Email Protocols Details]

A) Postfix / Sendmail

B) Google Chrome

C) FileZilla

D) Notepad

✓ **Answer: Postfix / Sendmail**

Q388. How does SSH authenticate users and servers without sending passwords in cleartext? [SSH Security Details]

A) Using public-key cryptography (asymmetric key pairs) or encrypted password hashes over a secure tunnel

B) By sending passwords in Base64 encoding

C) By relying on physical router passwords

D) By matching MAC addresses

✓ **Answer: Using public-key cryptography (asymmetric key pairs) or encrypted password hashes over a secure tunnel**

Q389. What tool included with SSH enables secure, encrypted file transfers between remote computers? [SSH Security Details]

A) SFTP (SSH File Transfer Protocol) and SCP (Secure Copy)

B) Telnet

C) Trivial FTP (TFTP)

D) Gopher

✓ **Answer: SFTP (SSH File Transfer Protocol) and SCP (Secure Copy)**

Q390. What is Port Forwarding (SSH Tunneling)? [SSH Security Details]

A) A mechanism in SSH that encrypts and routes network data from an arbitrary application through an established SSH connection

B) Modifying router firewall ports permanently

C) Opening physical server cases

D) Assigning public IP addresses to switches

✓ **Answer: A mechanism in SSH that encrypts and routes network data from an arbitrary application through an established SSH connection**

Q391. What is the Compression Ratio? [Data Compression Principles]

A) The ratio of the size of the uncompressed data to the size of the compressed data (Uncompressed Size / Compressed Size)

B) The time taken to download the file

C) The number of bits corrupted during transmission

D) The cost of storage disks

✓ **Answer: The ratio of the size of the uncompressed data to the size of the compressed data (Uncompressed Size / Compressed Size)**

Q392. What is Huffman Coding used in lossless compression? [Data Compression Principles]

A) An entropy encoding algorithm that assigns shorter variable-length bit codes to more frequently occurring symbols

B) An encryption cipher using prime numbers

C) A routing algorithm for packet switches

D) A method to calculate TCP checksums

✓ **Answer: An entropy encoding algorithm that assigns shorter variable-length bit codes to more frequently occurring symbols**

Q393. What is Run-Length Encoding (RLE)? [Data Compression Principles]

A) A simple lossless compression method that replaces consecutive repeated data values with a single value and a count

B) A method to run faster on fiber optic cables

C) A mechanism to increase packet transmission speed

D) An algorithm to detect network loops

✓ **Answer: A simple lossless compression method that replaces consecutive repeated data values with a single value and a count**

Q394. How would the text sequence 'AAAAABBBCC' be represented using basic Run-Length Encoding (RLE)?
[Data Compression Principles]

A) 5A3B2C

B) A5B3C2

C) 10ABC

D) AABBCC

✓ Answer: 5A3B2C

Q395. What is the trade-off of Lossy compression compared to Lossless compression? **[Data Compression Principles]**

A) Lossy yields much smaller file sizes and higher compression ratios at the expense of irreversible loss of subtle visual/audio detail

B) Lossy compression requires 100 times more storage space

C) Lossy files can never be opened on mobile devices

D) Lossy compression only works on plain text files

✓ Answer: Lossy yields much smaller file sizes and higher compression ratios at the expense of irreversible loss of subtle visual/audio detail

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